MILITARY SPECIFICATION

TEST PROGRAM SET (TPS) -

OPERATIONAL TEST PROGRAM SETS (OTPS)

FOR F/A-18 UNITS UNDER TEST (UUT),

GENERAL SPECIFICATION FOR

This specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines the general requirements for performance, design, development, fabrication, integration, test, and documentation, of operational test program sets (OTPS) and test program sets (TPS) used with the F/A-18 automatic test equipment (ATE), and preparation for delivery of OTPSs.

1.2 Intended use. The TPS as part of the OTPS will be utilized by Navy personnel of skill level E-4 or above at intermediate and depot level maintenance activities, to perform maintenance on selected units under test (UUTs).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6625

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1.3 Referenced paragraph citation. Where paragraphs of this specification or any of its referenced documents are cited, the citation shall be understood to include all subparagraphs unless otherwise noted.

2. APPLICABLE DOCUMENTS

2.1 Referenced Documents. The documents listed in Section 2, hereunder, form a part of this specification to the extent invoked by specific reference in other paragraphs of this specification. If a specification is referenced within this document without indicating any specific paragraphs as being applicable, then the specification is applicable in its entirety. Where a specific issue of the document is specified below, no other issue shall be used without the prior written approval of the Navy. When documents are referenced in other sections, a short form citing only the basic number of the document is used, revision letters, amendment indicators, notices, supplements and dates are omitted.

2.2 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-8421 - Air Transportability Requirements, General Specification for

MIL-N-18307 - Nomenclature and Identification for (NAVY) Electronic Aeronautical and Aeronautical Support Equipment Including Ground Support Equipment

MIL-P-116 - Preservation - Packaging, Methods of

MIL-T-28800 - Test Equipment for use with Electronic and Electrical Equipment, General Specification for

MIL-P-23377 - Primer Coating, Epoxy-Polyamide Chemical and Solvent Resistant for Weapons System

MIL-M-38784 - Manuals, Technical, General Requirements

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MIL-T-85549 (AS)

MIL-C-55302 - Connectors, Printed Circuit Sub-assembly and Accessories

MIL-C-81773 - Coating, Polyurethane, Aliphatic Weather Resistant

MIL-T-81914 - Tubing, Plastic, Flexible, Convoluted, Conduit, General Specification for

DOD-D-1000 - Drawings, Engineering and Associated Lists

STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-454 - Standard General Requirements for Electronic Equipment

MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment

MIL-STD-882 - System Safety Program for Systems and Associated Subsystems and Equipment, Requirements for

MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment for Facilities

MIL-STD-2076 - Unit Under Test Compatibility with Automatic Test Equipment, General Requirements for

MIL-STD-2077 - Test Program Sets, General Requirements for

MIL-STD-2084 (AS) - Maintainability of Avionic and Electronic Systems and Equipment, General Requirements for

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HANDBOOK

MILITARY

MIL-HDBK-217	- Reliability Stress and Failure Rate Data for Electronic Equipment

PUBLICATIONS

MILITARY

OPNAVINST 4790.2B	- The Naval Aviation Maintenance Program (NAMP) - Volumes I, II, III, and IV
NAEC-MISC-92-0555 (1 August 1979)	- Program Design Handbook for Automatic Test Equipment
NAEC-MISC-92-0554 (1 August 1979)	- Avionics Design Guide for ATE Compatibility
NAEC-MISC-92-0564 (26 March 1982)	- General Acceptance Test Procedure (GATP) for Test Program Sets (TPS) and Operational Test Program Sets (OTPS) for F/A-18 Units Under Test
PRD No. 39000013-001 (May 1981)	- Hybrid Test System (HTS), ATLAS User's Guide
PRD No. TBD	- Avionic Test Set (ATS), ATLAS User's Guide

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.3 Other publications. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI-STD-Y32.2	- Graphic Symbols for Electrical and Electronic Diagrams

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ANSI-STD-Y32.16A - Reference Designations for Electrical and Electronics Parts and Equipment

(Applications for copies should be addressed to the American Standards Institute, 1430 Broadway, New York, NY. 10018.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE-STD-416-1981 - IEEE Standard ATLAS Test Language

IEEE-STD-416A-1976 - ATLAS Syntax

IEEE-STD-771-1980 - Guide to the Use of ATLAS

(Application for copies should be addressed to the IEEE Standards Office, 345 East 47th Street, New York, NY. 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Item definition. The operational test program set (OTPS) is the result of merging one or more test program sets (TPSs) into a group which share a common interface device (ID) for the purpose of testing a selected reasonable grouping of weapon replaceable assemblies (WRAs) or shop replaceable assemblies (SRAs). In addition to the UUT TPSs comprising the OTPS, a self-test for the common ID shall also be included. The requirements of the ID self-test shall be as specified in 3.2.4.2.1. The TPS consists of the following elements

a. Test Program (TP)

b. Interface Device (ID)

c. Test Program Instruction (TPI)

d. Supplementary Data

3.1.1. Test program (TP) definition. The test program (TP) is a set of coded instructions residing on a

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test program medium (TPM) that controls the testing of the UUT. The objective of the TP shall be to automatically ascertain the operational readiness condition of the UUT, detect the presence of faults in the UUT, and isolate a fault to the required level specified herein. This is accomplished by ordered test routines which process stimulus and response data to and from the UUT, and evaluate the results. All test programs of an OTPS shall be merged onto a common TPM.

3.1.2 Interface device (ID) definition. The interface device (ID) provides the necessary electrical, mechanical, hydraulic, pneumatic, radiated, and optical interfaces between the ATE and the UUT. An ID can consist of simply a panel ID which mates directly with the UUT/ATE interface of the ATE, but can also include a cable set and test fixture as required by individual TPS requirements. All of the ID requirements of the individual TPSs of an OTPS shall be merged into a common ID.

3.1.2.1 Panel ID. The panel ID shall contain the necessary wiring and circuitry to interface the UUT to the ATE and to resolve any incompatibilities which exist between the ATE and the UUT in order to implement the test requirements. The panel ID shall provide electrical interface connectors to electrically interface the UUT operational connector(s) and test connector(s) via an external cable assembly. UUTs, which are plug-in circuit card assemblies, shall plug directly into circuit card receptacles mounted on the panel ID and shall be physically supported by the panel ID.

3.1.2.2 Cable set. The cable set, when required, shall provide the means to route power, stimulus, measurement and test point signals between the UUT/ID/ATE to effect testing of the UUT and self test of the panel ID, when applicable.

3.1.2.3 Test fixture. When required by a particular UUT, and in addition to the panel ID, a test fixture shall be provided as part of the TPS to provide an electrical and mechanical interface with the UUT. In such cases, the cable set interfaces the panel ID to the test fixture.

3.1.3. Test program instructions (TPI) definition. The test program instruction (TPI) shall provide TPS related information needed for testing, (e.g. hook-up, probe point locations, or other programmed operator intervention which cannot be conveniently provided or dis-

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played by the ATE under program control). The TPI shall be prepared in accordance with the format of Appendix A of MIL-STD-2077. All TPIs of an OTPS shall be merged into a common operational test program instruction (OTPI).

3.1.4. Supplementary data definition. The supplementary data consists of information, text, schematics, logic diagrams, and UUT source data necessary to fully develop, document, deliver, operate, repair and modify all elements of a TPS as well as analyze the TPS and UUT in the event of a problem or anomaly during the testing process. Supplementary data shall conform to the requirements of 3.3.4 herein. The supplementary data within an OTPS shall consist of the supplementary data packages of the individual TPSs within the OTPS.

3.2 Characteristics

3.2.1. Performance characteristics. Each TPS of an OTPS shall meet all requirements when operated in conjunction with the associated ATE and UUTs. The OTPS shall not be damaged by operation of the associated ATE and UUTs in any mode of operation (including OFF mode). The performance and capabilities of the associated ATE and UUTs shall in no way be degraded or damaged by operation of the equipment covered herein. All references to ATE shall be understood to mean both the test equipment hardware and applicable systems software.

3.2.1.1 Performance tests (PT). Performance tests (PT) shall contain all tests required to determine that a UUT is ready for issue (RFI). This shall be accomplished by discrete tests designed to automatically detect and annunciate the presence of any malfunction within the UUT that degrades system performance beyond its operational performance requirements and shall be performed without the use of ancilliary test equipment. Satisfactory completion of the performance tests shall assure that the UUT will perform its designed function when inserted into its next higher assembly. All internal power supplies and clocks shall be verified immediately following power application, if feasible. The performance tests shall be designed to detect those faults with a higher probability of occurrence early in the program to minimize program run time. The performance tests shall also test the built-in-test (BIT) circuitry of the UUT and cause the TP to do a proper fault isolation in the event a fault is detected. Where practicable, BIT information shall be used early in the performance tests to aid in reducing program run times. BIT shall not, however, be a substitute for any functional performance tests.

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3.2.1.1.1 UUT performance testing interface. Access to signals shall be through external operational and test connectors. Removal of UUT covers, use of extender boards, probing or any combination of the above, shall not be permitted during performance testing.

3.2.1.2 Fault detection. The TPS shall detect faults which degrade UUT performance to the extent that the designated functions of the UUT cannot be performed. Proper operation of BIT circuits shall be considered part of the UUT designated function. The UUT designated function(s) shall be derived from government furnished UUT source data.

3.2.1.3 Diagnostic fault isolation (DFI) tests. The diagnostic fault isolation (DFI) tests shall identify fault(s) detected by the performance test(s) and isolate to the next lower level of assembly. Fault isolation accuracy and limitations to ambiguity group sizes shall be in accordance with the specifications herein. It shall, however, be a design goal for all DFI routines to isolate to the next lower replaceable unit by automatic testing. The DFI shall consist of fault isolation routines designed to isolate to those faults with a higher probability of occurrence early in the diagnostic sequence to minimize program fault isolation run time. All testing results shall be displayed via the ATE cathode ray tube (CRT) and printer. When a repairable assembly is specified to be replaced, any additional information derived from these tests which may be beneficial in its repair, shall also be printed on the ATE printer. For example, when an ambiguity is resolved based on the failed condition of a component(s) in the specified replaceable unit, then that component(s) shall be called out as well.

3.2.1.3.1 UUT fault isolation testing interface. Access to signals shall be through external operational or test connectors. It shall be a design goal that removal of UUT covers use of extender boards probing, or any combination of the above shall not be necessary in order to meet the requirements of 3.2.1.3.2 during DFI.

3.2.1.3.2 Ambiguity groups. The order of priorities in determining ambiguity group sizes shall be as follows:

a. It shall be a design goal to fault isolate to a single replaceable unit.

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b. It shall be a design goal that the size of the ambiguity group shall be inversely related to the frequency of fault occurrence, i.e., high failure rate faults should be limited to small ambiguity groups.

c. As a minimum, the size of ambiguity groups shall meet the requirements of the following subparagraphs.

3.2.1.3.2.1 WRA fault isolation. Diagnostic WRA programs shall meet the following minimum ambiguity group requirements

a. For at least ninety (90%) percent of the detected faults of each SRA, the fault shall be isolated to that sole SRA.

b. For at least ninety-five (95%) percent of the detected faults of each SRA, the fault shall be isolated to that sole SRA and no more than one other SRA.

c. At a minimum, all detected faults of each SRA shall be isolated to that sole SRA and no more than two other SRAs.

d. All detected faults occurring in chassis mounted piece parts shall be isolated to that sole piece part.

3.2.1.3.2.2 SRA fault isolation. Diagnostic SRA programs shall meet the following minimum ambiguity group requirements

a. When the SRA contains ten (10) or fewer non-repairables, isolation to groups of two or less shall be possible for sixty (60%) percent of the detected faults. Isolation to four or less shall be possible for all detected faults. Each isolation group shall contain no more than one source integrated circuit (IC) package.

b. When the SRA contains more than ten (10) non-repairables, isolation shall be as follows:

1. Digital SRAs (ten percent or less of the active components are non-digital (linear)) -- Isolation to groups of four or less shall be possible for eighty-five (85%) percent of the detected faults.

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Isolation to groups of eight or less shall be possible for one-hundred (100%) percent of the detected faults. Each isolation group shall contain no more than three source IC packages.

2. Non-digital SRAs - Isolation to groups of four or less shall be possible for eighty (80%) percent of the detected faults. Isolation to groups of eight or less shall be possible for ninety-five (95%) percent of the detected faults. Isolation to groups of ten (10) or less shall be possible for one-hundred (100%) percent of the detected faults. Each isolation group shall contain no more than two source IC packages.

3.2.1.4 Adjustment/alignment routine. Adjustment/alignment routines are used for accommodating parameter variations within the UUT. A successful adjustment/alignment shall result in a program return to the performance test group at which the out of adjustment/alignment condition was detected. Failure to adjust shall result in an entrance into DFI.

3.2.1.5 ATE survey test. The ATE survey test shall be designed so as to maintain ATE system confidence between periodic ATE system self-test runs and to ensure that each UUT test will provide meaningful results. This shall be accomplished by routines designed to test the ATE resources for the unique stimulus and measurement requirements of the TP when testing a specific UUT. The ATE resources shall be tested to the accuracies required by the UUT. The ATE survey test is a part of the test program of the individual TPS and may be performed or bypassed based on ATE operator options.

3.2.1.6 Identity (signature) tests. Each TPS shall include signature tests so as to verify that the UUT/ID combination and their interconnection is correct. The tests are intended to preclude equipment damage and, thus, should not themselves cause damage. These tests shall be the first tests executed by the test program.

3.2.1.7 Safe-to-turn-on (STTO) test. The safe-to-turn-on-test(s) shall be provided so as to preclude conditions that could damage the UUT, ID, or ATE when power is applied to the UUT/ID. The test(s) shall not be restricted to UUT/ID power lines, signal lines must also be considered.

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3.2.1.8 ID confidence test. An ID confidence test shall be implemented for those UUT TPSs which utilize ID circuitry containing active devices. The TPS confidence test shall verify the operational capability of that portion of the ID which is pertinent to the individual TPS. The ID confidence test shall be performed with the UUT connected to the ID, where feasible. When a connected UUT precludes a valid test of the ID active circuitry required for testing that UUT, then the UUT shall be removed during the execution of the confidence test. In the event that a fault is detected, automatic fault isolation to levels in accordance with 3.2.1.3.2 shall be performed. The ID confidence test execution time shall not exceed five (5) minutes. The station operator shall have the option of bypassing the ID confidence test.

3.2.2 Physical characteristics. The TPS shall provide the following physical characteristics in addition to being physically compatible with the targeted ATE test station.

3.2.2.1 Interface device (ID). As the hardware element of the OTPS, the ID shall provide the physical and electrical interface required to implement testing of the UUT, and shall resolve any incompatibilities between the ATE and UUT.

3.2.2.1.1 Electrical. Electrical incompatibilities between the UUT and ATE shall be resolved by the use of active circuitry, passive circuitry, or both in the ID. The use of passive and active circuitry shall be subject to the requirements of 3.2.2.1.3. The electrical design of the ID shall not utilize circuitry requiring alignments or adjustments.

3.2.2.1.2 Test point interface. The ID design shall include the necessary access to the external interface of the UUT and to the UUT test points.

3.2.2.1.3 ID complexity. It is a requirement to minimize the ID complexity by using a design and configuration as simple and with as few parts as possible. Passive circuitry (other than straight-through wiring) may be used only when the capability is not otherwise provided by the ATE. Active circuitry may be used only when the capability is not otherwise provided by the ATE, passive circuitry, or a combination of the ATE and passive circuitry.

3.2.2.1.4 ID optimization. The IDs shall be optimized so that as many UUTs as practical can be tested by the same ID with the objective of reducing the total number of IDs required.

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3.2.2.1.5 ID expansion. There shall be a twenty (20%) percent expansion capability in the design of the ID to accommodate unanticipated or future ID requirements including number of wires, added functions, and subassemblies.

3.2.2.1.6 ID size and weight. The ID shall meet the following requirements

a. Size and weight must be compatible with the targeted ATE work surface.

b. It must be designed so that one man can mount the ID onto the ATE interface connector.

c. The weight of that portion of the ID including cables which must be fastened to the ATE interface connector shall not exceed thirty (30) pounds. The total weight of an ID shall not exceed the requirements of MIL-STD-1472.

d. IDs shall be designed in conformance with the requirements of MIL-T-28800 type III class 4 equipment.

3.2.2.1.7 Signature testing. Each ID shall contain signature resistor(s) for ID signature tests. Signature resistor values shall be assigned by the Navy.

3.2.2.1.8 ID accessibility. ID test and maintenance access shall be provided on the ID surface which promotes the greatest amount of accessibility. Entry into the ID through the access covers shall provide access to all interface wiring. With the ID connected to the ATE interface, it shall be possible to extend ID SRAs, and to test plug-in ICs.

3.2.2.1.9 ID enclosure. The ID enclosure shall be designed to meet the requirements of the following subparagraphs.

3.2.2.1.9.1 Drip-proof. The ID enclosure need not be a drip proof design but shall meet the requirements for an open protected enclosure as specified in MIL-STD-108. The enclosure shall be an aluminum weldment with the access covers specified in 3.2.2.1.8 herein.

3.2.2.1.9.2 Feet. Feet shall not be included on the ID enclosure.

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3.2.2.1.9.3 Handles. Two fixed handles, shall be installed on the ID. The handles shall be mounted on the front panel with one handle located approximately one inch from each vertical side of the front panel. The handles shall be oriented so the grip will be vertical when the ID is mated with the ATE interface connector, and of sufficient height to protect all front panel connectors and other hardware.

3.2.2.1.9.4 External connectors. The ATE interface connectors shall be on the panel ID back plane. The ID cable and UUT interface cable connectors shall be located on the panel ID front surface, and shall be in accordance with 3.3.2.4.2.

3.2.2.1.9.5 Attachment of nameplate. The identification nameplate of the ID shall be firmly attached to the panel.

3.2.2.1.9.6 Chassis fastening. The ID chassis shall be securely fastened within the ID case by the use of screws or positive, rapid engagement fasteners.

3.2.2.1.10 Test set-up time. The OTPS shall be designed such that the UUT hook-up to the ATE and pre-UUT performance test activities are minimized and do not exceed ten (10) minutes.

3.2.2.2 Useful life. The ID shall have a useful life of not less than ten years under any combination of operating and storage life without exceeding the operational service life.

3.2.2.2.1 Operational service life. Tne ID shall have an operating life of not less than ten thousand (10,000) hours under the operating environmental conditions specified in 3.2.5 with normal preventive maintenance.

3.2.2.2.2 Storage life. The ID shall have a storage life of up to thirty (30) months under the environmental service conditions specified in 3.2.5 without parts replacement or unscheduled maintenance. The ID, upon maintenance shall be capable of meeting the requirements of this specification for no less than the total operational service life.

3.2.3 Reliability. Equipment reliability shall be considered in every phase of the design and development process and shall be in accordance with best commercial practices.

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3.2.3.1 ID mean-time-between-failure (MTBF). Each ID shall have a minimum mean-time-between-failure (MTBF) of 2500 hours calculated in accordance with MIL-HDBK-217.

3.2.4 Maintainability. The maintainability requirements of the TPS shall be as described in the following subparagraphs.

3.2.4.1 Test program maintainability. The following provisions shall be accommodated to aid in maintaining the test program.

a. There shall be a cross reference in the test program listing between the test numbers and the appropriate flow chart for all tests.

b. Test program test numbers shall be incremented at least by tens to permit inclusion of program changes.

c. The documentation defined in 3.4 shall be used as reference information when modifying the program.

3.2.4.2 ID maintainability. The ID shall be designed to the requirements contained in MIL-STD-2084, MIL-STD-2076, NAEC-MISC-92-0554, "Type III maintainability" of MIL-T-28800, and "Design for maintainability ofMIL-STD-1472. The equipment shall also be capable of being maintained as defined in OPNAVINST 4790.2B. Design of the TPS and OTPS shall emphasize simplicity in operation and be free from unscheduled maintenance to the maximum extent possible. The TPS and OTPS shall be designed such that calibration and scheduled maintenance shall not be required.

3.2.4.2.1 ID self-test. The ID self-test TPS shall ascertain the operational readiness of all the circuitry in that ID. In the event of fault detection, it shall automatically isolate the fault, which includes active and passive circuitry, wiring, etc., to the required levels of 3.2.1.3.2. The ID self-test TPS shall be designed to meet all the requirements of a UUT TPS as specified herein, and shall test the ID on the ATE in the same manner as a UUT, however, no additional ID shall be required. ID self-test shall be performed with the UUT disconnected from the ID and with the UUT interface wrapped around to the ATE via shorting plugs and cables which, if

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practicable, shall be stored in the ID. Shorting plugs and cables in this context are not considered IDs. The ID self-test TP shall be encoded and stored on each test program medium with which that particular ID is used.

3.2.4.2.2 ID mean-time-to-repair (MTTR). The ID MTTR which includes the ID confidence test run time and repair time shall not exceed thirty (30) minutes.

3.2.5 Service conditions (environmental). The ID shall not suffer damage, deterioration or degradation of performance beyond the limits of this specification when subjected to the extremes of environmental service conditions specified herein.

3.2.5.1 Temperature-altitude combination. The ID shall meet the performance requirements of this specification under any possible combination of the temperature altitude condition shown in Table I. A twenty (20) minute warmup period is allowed prior to operation after being subjected to the temperature extremes specified in Table I.

TABLE I. Temperature-altitude

Equipment operating			Equipment non-operating	
Temperature extremes °C				
Continuous	Intermittent 20 minutes	Altitude	Temperature extremes °C	Maximum altitude
0 to +40	+50	0 to 6,000 ft. (30.0 to 24.4 in Hg)	-40 to +75	50,000 ft. (3.4 in Hg)

3.2.5.2 Humidity. The ID shall meet the performance requirements of this specification under humidity conditions up to 95% relative humidity. Operation under conditions wherein condensation occurs in the ID is not required.

3.2.5.3 Vibration. The ID shall meet the performance requirements of this specification under the vibration conditions specified below.

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TABLE II. Vibration conditions

Frequency Range (Hertz)	Displacement ------ Double Amplitude (inches)	
	Operating	Non-Operating
4-15	0.030 ± 0.006	0.060 ±0.006
16-25	0.020 ± 0.004	0.040 ±0.004
26-35	0.010 ± 0.002	0.020 ±0.002
36-40	0.005 ± 0.001	
41-50	0.003 + 0.000 - 0.001	

3.2.6 Transportability. In addition to meeting the transportability requirements of MIL-A-8421, the TPS, when packaged for shipment shall be transportable by all modes of transportation to permit employment, deployment, and logistic support. The equipment shall provide ease in handling, loading, and securing.

3.3 Design and construction.

3.3.1 Test program. The test program shall determine if the UUT is operating within its operational limits and shall also determine which sub-element of the UUT has failed if the UUT fails a test. To implement this function the applicable methods, techniques, procedures, etc., described in the ATLAS User's Guide of the designated ATE, the Program Design Handbook for Automatic Test Equipment, and the IEEE Guide to the Use of ATLAS shall be used. The precedence for the use of these documents shall be in the order listed, except that the requirements as stated in this specification shall have the highest precedence. The test program shall perform its function by testing the UUT in accordance with the following requirements.

3.3.1.1 Test program language. The individual test programs shall be prepared using the ATLAS source language of IEEE-STD-416-1981 and IEEE-STD-416A-1976 as definitized by the ATLAS language defined in the ATLAS User's Guide of the designated ATE.

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3.3.1.1.1 Computer program definition requirement. The test program shall operate in conjunction with the ATLAS test executive to control the application of stimuli from the ATE to the UUT and to monitor, analyze and evaluate the responses from the UUT.

3.3.1.2 Required UUT documentation. The individual test programs shall be written in ATLAS and shall be based on UUT source data.

3.3.1.3 Test program sequence. Test programs shall be written in accordance with the following standard sequence.

3.3.1.3.1 Identity (signature) tests. The test program shall commence with ID and UUT identification tests as described in 3.2.1.6. Failure of these tests shall result in a message to the operator identifying the failure and termination point of the test.

3.3.1.3.2 ATE survey test. The ATE survey test as described in 3.2.1.5 shall, if selected, be run before the UUT tests.

3.3.1.3.3 Safe-to-turn-on (STTO) tests. The STTO tests as defined in 3.2.1.7 shall be performed before application of power.

3.3.1.3.4 Power application. Power shall be applied to the UUT/ID in accordance with the UUT/ID requirements after successful completion of STTO as defined in 3.2.1.7. When necessary, the TP shall incorporate time delays to make suitable provision for UUT warm-up.

3.3.1.3.5 ID confidence test. The ID confidence test as described in 3.2.1.8 shall, if selected, be run before the UUT tests.

3.3.1.3.6 Performance test. The UUT performance tests (end-to-end tests) as defined in 3.2.1.1 shall determine if the UUT is ready-for-issue (RFI). Performance tests shall sequence via "GO" paths. "NO-GO" paths shall either lead to diagnostic tests, adjustment/alignment tests, or to a UUT failure message.

3.3.1.3.7 Diagnostic tests. Diagnostic tests as described in 3.2.1.3, shall be performed upon a failure of PT and shall isolate to replaceable units.

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3.3.1.3.8 Adjustment/alignment tests. Where applicable, adjustment/alignment tests shall be performed upon a failure of PT as per 3.2.1.4. Improper operator actions, such as misadjustment of a UUT, shall be anticipated by allowing a second chance in the event of an erroneous test result.

3.3.1.3.9 Removal of UUT power and stimuli. Once testing has been completed through either fault identification or successful completion of the program, all power, stimuli and measurement device connections shall be removed from the UUT and ATE interface under program control. All power connections should be removed prior to any message requiring operator participation in the shut down procedure. Discharge paths shall be provided as required.

3.3.1.4 ATE resources. Where feasible, the test program shall be designed to utilize ATE station resources. The use of ID circuits shall be consistent with the requirements of 3.2.2.1.3.

3.3.1.5 Test cycling. Where feasible, the test program shall be structured to permit the ATE to cycle individual tests. That is, in order for the measurement associated with the test to be valid, the test by its nature must have its required stimulus either applied in the test or in a previous test.

3.3.1.6 Single/multiple fault criteria. Diagnostic tests shall be written on the premise that no more than one fault exists within each independent function of the UUT.

3.3.1.7 Tolerance coning. The test program design shall incorporate test limits ensuring that when a UUT is installed in its next higher level of assembly it shall allow that assembly to operate in accordance with its performance limits.

3.3.1.7.1 Accuracy requirements. In order to assure sufficient TP accuracy with respect to the UUT limits, as well as reliability and repeatability of test results, TPs shall be designed to meet minimum test accuracy ratios (TARs) of three-to-one (3 1).

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3.3.1.8 Automatic test program generator (ATPG). Where practicable, an automatic test program generator (ATPG) shall be used during the design of a test program, and it shall be of a Navy approved type. The ATPG programs require the UUT circuit to be modeled in accordance with the format specified by the identified ATPG. If an ATPG is utilized, dynamic failure modes must still be considered, and if necessary, manually generated tests shall be included. Further, manual test design techniques shall be incorporated, if necessary, to meet the fault detection and fault isolation requirements of this specification when utilizing ATPG.

3.3.1.9 Test program elements. The hierarchy of elements of the test program shall consist of program partitions, test groups, tests, and statements. Specific descriptions of each constituent are provided in the following subparagraphs. The test program shall consist of one or more performance/diagnostic partitions.

3.3.1.9.1 Program partition. The test program for a given UUT shall be segmented into one or more stand-alone test partitions as described in the designated ATE ATLAS User's Guide. Each partition shall be limited to a maximum of 2000 ATLAS statements, and shall be capable of being compiled, debugged and integrated on the designated ATE as an entity. The procedural section of the partition, as described in the designated ATE ATLAS User's Guide, shall consist of one or more performance test groups and its related diagnostic test groups (performance/diagnostic partition). Each partition shall commence with the performance test group(s) and shall be followed by the diagnostic test group(s) in their entirety. A partition shall have one or more entry point(s) subject to the rules stated in the following subparagraphs. The content of each partition shall be in accordance with this specification.

3.3.1.9.1.1 Entry points. Entry points are indicated by an Entry point flag (ATLAS E flag) and are addressable points in the test program which may be entered without regard to previously established ATE settings. Start of test program execution shall be possible from any entry point providing that the requirements of 3.2.1.6 and 3.2.1.7 are not violated.

3.3.1.9.1.1.1 Initial entry point. The initial entry point shall be the first entry point of the first partition and shall initiate the test program sequence as defined in 3.3.1.3.

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3.3.1.9.1.1.2 Performance entry point. A performance entry point shall consist of an entry point flag, protection tests (signature and safe-to-turn-on tests) and standard UUT test setups including power-up as needed. A performance entry point shall be used to begin every performance test group. During normal execution of the test program, i.e., from the initial entry point, the performance entry point's protection tests and test setups already in effect shall be bypassed. When a program is started at a performance entry point, program flow after completion of the protection tests and test setups shall be the same as for normal execution.

3.3.1.9.2 Test group. A test group shall consist of one or more tests which are to be executed in a logical order determined by statements contained within the group. There shall be two types of test groups, - performance test groups and diagnostic test groups - distinguished by the type of testing contained therein and by a requirement that only performance test groups have entry points. The size of a test group shall be subject to the following criteria. (1) A performance test group shall have a maximum size of 500 ATLAS statements as a design goal. (2) A performance test group shall, as a design goal, execute in ten minutes or less. (3) The tests constituting a performance test group shall be related by UUT circuit functions. (4) A diagnostic test group shall not exceed the maximum partition size.

3.3.1.9.2.1 Performance test groups. Performance test groups consist of one or more performance tests. Each performance group shall have one entry point at the beginning of the group permitting entry into the first test of that group. A fault detection within the performance test group, shall cause the test program to automatically enter an adjustment/alignment routine or a diagnostic test group and to display proper instructions to the operator via the ATE CRT.

3.3.1.9.2.1.1 Test execution integrity check. A test execution integrity check shall be implemented in the test programs to preclude the classification of a UUT as RFI if all performance test groups have not been executed due to operator use of ATE test execution control options.

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3.3.1.9.2.2 Diagnostic test group. A diagnostic test group shall consist of the diagnostic tests which implement fault isolation, following the detection of a UUT failure by a particular performance test or the inability to successfully adjust the UUT via an adjustment/alignment routine. Diagnostic test groups shall not have entry points. All branches of a diagnostic test group shall ultimately terminate in a repair action message.

3.3.1.9.3 Test. A test, as a minimum, shall consist of a measurement and a decision based on the result of the measurement(s). Depending on the decision reached, test program execution shall proceed from a performance test, to (1) the next performance test, (2) an adjustment/alignment routine, (3) a diagnostic fault isolation test or (4) a replace action. Test program execution shall proceed from a diagnostic fault isolation test to, (1) another fault isolation test, (2) an adjustment/alignment routine, or (3) a replace action.

3.3.1.9.4 Statement. The statement is the lowest level programming constituent referenced within this specification and, except for comment statements, generally consists of a flag field, statement number consisting of test, step, or test and step numbers, verb, statement body and statement terminator (dollar sign ($)). The syntax of test program statements is defined in the designated ATE ATLAS User's Guide.

3.3.1.10 Program heading and identification. Program heading and identification data shall contain the test program name, identification number, and revision number. It shall also include the ID identification number. The format for program heading and identification shall be consistent for all TPSs and OTPSs. The test program and its partitions shall each be uniquely identified in accordance with the following subparagraphs.

3.3.1.10.1 Test program identifier. The test program and its configuration shall be identified by a test program identifier. This identifier shall be used as part of the partition name.

3.3.1.10.2 Partition name. A partition name shall consist of the test program identifier and a two digit partition number. Partition numbers shall be assigned such that their numeric order reflects the order of UUT test program execution.

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3.3.1.10.3 UUT identification. Comment lines near the beginning of each test program partition shall be used to identify the tested UUT. The UUT identification data shall consist of UUT part number and UUT nomenclature and shall be displayed via the ATE CRT and printer.

3.3.1.10.4 ID identification. Comment lines near the beginning of each test program partition shall be used to identify, by their alphanumeric part numbers, the ID with which the test program is used. This information shall be displayed via the ATE CRT.

3.3.1.11 Structured programming. The test program shall use ATE ATLAS structured statements (i.e. IF-THEN-ELSE, WHILE-THEN, FOR-THEN, etc.) as defined by the targeted ATE ATLAS User's Guide. The ATLAS GO TO statement shall be used only to unconditionally branch from a performance test to its related diagnostic test.

3.3.1.12 Test program coding format. The coding of the test program shall follow the format rules defined in the following subparagraphs. Their purpose is to make the test program clear and understandable.

3.3.1.12.1 Indentation rules. Indentation shall be used to allow the reader of the test program listing to scan the path of program flow and efficiently extract information. Indentation shall be required when ATLAS statements have continuation lines and when the ATLAS IF-THEN-ELSE, WHILE-THEN, FOR-THEN and DEFINE-PROCEDURE programming structures are used. Columns 1-8 of each line of test program source code shall be used only for the ATLAS flag, statement number and mandatory blank fields. (Columns 2-8 shall not be used for comment statement text even when permitted by the specific ATE's ATLAS). All program text shall be indented to one of the following tab positions columns 9, 13, 17, 21, 24, 27, 30, 32, 34, 35, 36, 37, etc. The tab position at column 9 shall be used for only comment text. All other text shall begin in column 13 or beyond. The first non-comment statement of a test program partition shall begin in column 13.

3.3.1.12.1.1 Comment indentation. The text of all ATLAS comment statements shall begin in column 9. All multi-line comments shall be written as a single ATLAS statement. The continuation lines of a comment statement shall be indented only to present tabular information or lists.

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3.3.1.12.1.2 Statement indentation. The text of all ATLAS statements which are not comment statements shall begin at the same tab position as the preceding non-comment statements unless the statement follows an ATLAS IF, ELSE, WHILE, FOR or DEFINE statement. The continuation lines of all non-comment statements shall begin one tab position to the right of the tab position of the first line of the statement.

3.3.1.12.1.3 Indentation within ATLAS programming structures. The tab position of the IF, ELSE, WHILE, FOR and DEFINE-PROCEDURE programming structures shall be the same as the beginning tab position of the preceding non-comment statement. The END statement for each of these four programming structures shall begin in the same tab position as the corresponding IF, WHILE, FOR or DEFINE keyword. All statements between an IF keyword and a corresponding ELSE keyword or between an IF, ELSE, WHILE, FOR or DEFINE keyword and a corresponding END statement shall begin one tab position to the right of the keyword. The IF-THEN-ELSE, WHILE-THEN and FOR-THEN programming structures may be nested within each other or within a DEFINE PROCEDURE structure. Statements governed by a nested programming structure shall be indented with respect to the keywords of the nested structure. Thus each time a structure is nested within an outer structure, the indentation level shall move one tab position to the right. The indentation rules are illustrated in the following example

```
C       TEST POWER SUPPLY
        $
 010010     APPLY, AC SIGNAL, VOLTAGE 115 V
                FREQ 400 HZ,
                CNX HI J1-15 LO J1-20  $
            .
            .

 010210     WHILE, 'INDEX' LT 30 , THEN $
     12         CALCULATE, 'ARRAY' ('INDEX') = 0.1    $
     14         CALCULATE, 'INDEX' = 'INDEX' + 1    $
     16     END, WHILE $
            .
            .
```


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```
010412      IF, 'MEASUREMENT' GT 5.0 V , THEN  $
     14         IF, 'TEST' EQ 1, THEN $
     16             DISPLAY, MESSAGE, 'FAILURE'
                       C 'JUMPER J2-50 TO J2-1 ON 1A12' $
     18             FINISH   $
     20         ELSE  $
     22             DISPLAY, MESSAGE, 'FAILURE',
                       C'1A10'   $
     24             FINISH  $
     26         END, IF  $
     28      END, IF  $
                .
                .
                .
```

3.3.1.12.2 Comment usage. Comments shall be used throughout the test program to describe briefly and clearly the program and its subdivisions. Comments shall be used at predetermined locations within the test program as specified in detail in the following subparagraphs.

3.3.1.12.2.1 Program description. Comments at the beginning of each partition shall describe test program applicability (UUT/ID description), test program partitioning and test number assignments.

3.3.1.12.2.2 Section separation. Comment statements shall be used to separate the main sections of a test program partition. These main sections shall include the implementation structure, preamble and procedural sections. In a performance/diagnostic partition a section separator shall also be used between the performance test group(s) and the diagnostic test group(s). Each of these section separators shall consist of seven or more lines of comments, including a border of asterisks, for example

```
C
 ************************************************************
 *                                                          *
 *                    PREAMBLE SECTION                      *
 *                                                          *
 ************************************************************
 $
```

3.3.1.12.2.3 Test group and procedure delineator. Each procedural section test group and each ATLAS procedure definition function definition message definition and group of variable declarations in the preamble section shall be delineated with a series of seven or more lines of comment in the same manner as described above for section separators. The main difference is that the border of asterisks shall be adjusted to just accommodate the text within it. The ATLAS label assigned to a procedure, function or message shall be descriptive thereof

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and shall be included within the delineator for the procedure, function or message. The ATLAS labels assigned to variables shall be descriptive of the data represented by the variables. Groups of variable declarations shall be delineated as follows

```
C
        ************************
        *                      *
        *  DECLARE VARIABLES   *
        *                      *
        ************************
$
```

3.3.1.12.2.3.1 Test group delineator. The test group delineator shall describe the test group function, for example

```
C
        ************************
        *                      *
        * TEST POWER SUPPLIES  *
        *                      *
        ************************
$
```

3.3.1.12.2.3.2 Comment lines. The delineator shall be followed by a series of comment lines describing the function to be performed in more detail.

3.3.1.12.2.4 Test body comments. Each test group and each test within the group shall be preceded by comments briefly and concisely describing its purpose and any special conditions. The description shall include but not be limited to the UUT connector/pin designation, signal name, and the state of the signal independent of signal conditioning by the ID. The annotation of ATPG programs shall apply only to the non-ATPG sections. Comments shall be made visible by spacing from the preceding and following non-comment statements. Comments within a test shall be avoided unless the information can not be described at the test level.

3.3.1.12.3 Test numbering. Each program statement number shall consist of a six-digit number which will be further subdivided into a four-digit test number followed by a two-digit step number. As a design goal, preamble statements shall be assigned test numbers from 1 to 499; performance tests, numbers from 500 to 4999; and diagnostic tests, numbers from 5000 to 9999. Further, diagnostic test numbers shall be offset by an increment of 5000 from the invoking performance test number.

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3.3.1.13 Messages. The instructions, data, and information, which are needed by the operator to successfully initiate and complete the checkout of the UUT using the OTPS components, shall be communicated to the operator via the ATE CRT display. Messages which report the detection of a UUT performance failure, diagnosis of a failure, a repair action, or a determination that the UUT is good, shall be communicated to the operator via the ATE CRT display and ATE printer.

3.3.1.13.1 Standard messages. Appendix D of MIL-STD-2077 defines standard display messages. These standard messages shall be used within the test program as applicable. In addition to the requirements of MIL-STD-2077, messages signifying the end of the program as a result of a faulty UUT shall also include the nearest entry point preceding the performance test in which the fault was detected.

3.3.1.13.2. Non-standard messages. Non-standard messages, abbreviations or special mnemonics shall be identified to the Navy. These messages shall use generally accepted colloquial English language.

3.3.1.14 Test efficiency. Test efficiency shall be considered in the test program design. This shall include but not be limited to minimization of operator actions and run time. Run time shall be minimized by testing the least reliable lower order of assemblies of the UUT at the earliest feasible stages of the program.

3.3.1.15 Self-containedness. The tests shall be designed to operate with the designated ATE test station and require no external equipment or software.

3.3.1.16 Test operator interface. Communication with the operator shall consist of standardized messages as per 3.3.1.13.1 and displayed on the ATE CRT, printer or both. Messages which cannot be displayed on the CRT or which require amplification shall be contained in the test program instruction (TPI).

3.3.2 ID design and construction. The mechanical and electrical design of the ID shall be in accordance with best commercial practices and the following requirements. The design shall be compatible with the targeted ATE requirements, the OTPS intended use requirements as specified in 1.2, and shall be in accordance with the reliability requirements of 3.2.3.

3.3.2.1 Parts and materials. In the selection of parts and materials, fulfillment of major design objectives shall be a prime consideration. Proprietary parts and materials shall not be used. Part and material requirements shall conform to best commercial practices, shall be state-of-the-art, currently manufactured and of proven

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design such that performance, reliability and accuracy are readily verifiable. Spare parts shall be commercially available.

3.3.2.2 Design stresses. The ID shall be of adequate strength to withstand, without damage, stresses incident to moving and handling.

3.3.2.3 Cables. Cable assemblies shall be flexible, rugged, and constructed so that they can provide the necessary performance capability and electromagnetic interference (EMI) protection as required in 3.3.2.5, and can be easily maintained. Individual signal wire shields, if required for EMI protection, shall be in accordance with 3.3.2.16 and MIL-T-28800, "Shielded cables." The EMI shields shall be integral to the outer covering of the cable and shall be terminated in fittings that mate with or are part of the connector backshell adapters. These adapters shall be removable and reuseable and shall provide EMI protection. Cables shall be protected by external tubing (convoluted) per MIL-T-81914. The external tubing shall contain, as an integral part, an internal braided EMI shield. This shield shall be grounded peripherally to the connector backshell adapter. Cable connectors shall not be potted. Cable ties shall not be used on the wire bundle in convoluted tubing. Individual wires shall be of sufficient length to prevent damage when the cable is bent 180 degrees. Cables shall be constructed such that a minimum of three inches of wire is exposed between the cable covering and the connector when the connector backshell is removed. Cable assemblies shall contain no electrical components.

3.3.2.3.1 Strain relief. Strain relief provisions shall be incorporated on the cable assembly to protect cables and wires from breakage. Strain relief measures shall also be provided when the weight of a connector or a cable and connector can damage the connector on the UUT or the ID.

3.3.2.3.2 Coding. Coding of wires in accordance with MIL-T-28800, "Coding," is required for equipment furnished under this design specification.

3.3.2.3.3 Cable assembly configuration. Each cable assembly shall be an unbranched cable terminated on each end with a connector in accordance with 3.3.2.4.

3.3.2.4 Electrical connectors. ID connectors shall be compatible with the UUT and the ATE connectors. The ID connectors shall be of the following types.

a. Hughes MRS series shall be used for all ID panel connections. Adapter connectors shall not be used with the exception of unique connectors such as coaxial and waveguide connectors.

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b. Printed circuit board connectors shall be in accordance with MIL-C-55302.

3.3.2.4.1 Contacts. ID multiple pin connectors carrying power from the ID to the UUT shall have female-type contacts whereas ID multiple pin connectors receiving power from the UUT shall have male-type contacts.

3.3.2.4.2 Spacing and quantity. Connectors shall be located such that disconnecting any one connector does not require disconnecting any adjacent connector(s). If two or more UUT's associated with an OTPS require identical ID connector(s), then these connector(s) shall not be duplicated on the ID, i.e., for each UUT requiring them, common ID connector(s) shall be utilized.

3.3.2.4.3 Protective caps. All connectors on the ID shall be provided with protective caps and chain retainers. The retainer chains shall be covered by sleeving and in no way interfere with removal or replacement of the cap on the connector. The protective caps and chain retainers shall not interfere with any panel readings, controls, etc., during the operation of the ID.

3.3.2.4.4 Allowable internal ID connector types. Internal ID operational connectors shall be MIL-C-55302 type. Standardization of one size connector is required. Allowable connector types are:

a. M55302/27 - (92 or 98) style 2
b. M55302/58 - (40) - Y

3.3.2.4.5 Connector keying. Connectors of the same size within the ID shall be keyed/polarized to preclude mismating. Keying shall be accomplished in a manner that will allow all SRAs using the same connector to mate an unkeyed single test receptacle.

3.3.2.5 Electromagnetic interference control (EMIC). The ID shall operate satisfactorily in a Navy maintenance activity environment, and shall not degrade nor be degraded by the performance of any other adjacent similar support equipment installed in that environment. The ID shall be designed to comply with CE03, CS02, and RE02 of MIL-STD-461.

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3.3.2.6 Maintenance of electronic assemblies. To facilitate maintenance, ID construction shall provide access to electronic assemblies or be capable of operating with the electronic assemblies or subassemblies placed in extender assemblies. If extender assemblies are required, they shall be provided as part of the ID.

3.3.2.6.1 Printed circuit cards. All printed cards shall be identical in size.

3.3.2.6.2 Active devices. Active devices when approved for use in the ID, shall be located in sockets on plug-in printed circuit cards.

3.3.2.7 Standard buffering devices. For the purpose of interfacing UUT test points to the ATE interface, a mix of preferred standard microelectronic circuit buffers has been selected to be used in the ID if required. These consist of line drivers and receivers for analog and digital circuits. A list of these devices is contained in Appendix A herein.

3.3.2 8 Terminal blocks. When many connections must be tied together within the ID, pin type terminal blocks shall be used to facilitate easy repair of wiring.

3.3.2.9 Interchangeability. Equipment furnished as part of the ID, including major subassemblies and circuit cards, shall be interchangeable with other equipment of like part number and dash number, with respect to form, fit, and function.

3.3.2.10 Electrical identification. Each ID shall contain electrical circuits for the purpose of identification (signature) tests as per 3.2.1.6.

3.3.2.11 Finishes. The equipment shall be finished in accordance with MIL-T-28800, "Finishes, enclosure", and the following subparagraphs.

3.3.2.11.1 Protective finish. Exterior or exposed surfaces shall have a protective finish of one coat of MIL-P-23377 primer plus two coats of MIL-C-81773 polyurethane enamel.

3.3.2.11.2 Color. Color of the painted surfaces of the ID shall be green (color number 14260) in accordance with FED-STD-595. Cables shall be black tubing per MIL-T-81914. Interior colors shall be that resulting from the use of MIL-P-23377 primer.

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3.3.2.11.3 Lacquer. The use of lacquer is prohibited.

3.3.2.12 Nameplates and product markings. Unless otherwise specified, product markings shall be in accordance with MIL-T-28800, "Type III marking and identification," and the following

a. Nameplates shall be in accordance with the format to be supplied by the Navy.

b. The ID nameplates shall be located in the upper right corner of the front panel approximately 0.5 inch from each edge.

c. The MIL-T-28800, "Receptacles", requirements shall be modified so that all connectors shall be identified by sequentially numbered "J" or "P" reference designators. Mating connectors shall have matching reference designators i.e., J1 shall mate with P1. Connector mating information shall be provided when space permits.

d. Cable identification shall be in accordance with format to be supplied by the Navy.

e. Non-metallic bands shall be provided on each cable for identification of the cable. Except for multi-purpose cables, non-metallic bands shall also be provided for identification of each cable connector with which it mates.

f. All parts or assemblies which require special care during handling, shipping, installation or test shall be identified and marked.

g. Devices within the ID which are sensitive to electrostatic discharge shall be labeled as such with a decal.

h. MIL-T-28800, "Diagram, schematics, and special instruction sheet", requirements do not apply to the equipment furnished under this specification.

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3.3.2.12.1 Marking of nontype designated units, parts and assemblies. Nontype designated units, parts, and assemblies shall be marked in accordance with 3.4 and 3.9.7 of MIL-N-18307.

3.3.2.12.2 Identification of type designated units. The nameplate conforming to the supplied format shall be permanently attached to each type designated unit of equipment. Nomenclature for type designated units will be supplied by the Navy. Serial numbers shall be assigned by the Navy.

3.3.2.12.3 Identification of type designated sets and groups. The nameplate conforming to the Navy supplied format shall be permanently attached to the mounting structure of each type designated set or group of equipment. Nomenclature for type designated sets or groups will be supplied by the Navy. Serial numbers shall be assigned by the Navy.

3.3.2.13 Workmanship. Workmanship shall be performed in accordance with best commercial practices.

3.3.2.14 Safety. Safety shall be in accordance with 3.3.8.

3.3.2.15 Human engineering/human performance. Human engineering shall be in accordance with MIL-STD-454, requirement 621.

3.3.2.16 Grounding. Ground loops and common ground returns for signal and power grounds shall be avoided. The design of the ground systems shall protect operating and maintenance personnel from electrical hazards, provide the necessary shielding for signal circuits, and minimize EMI. Primary ac power returns (neutral) shall remain isolated from all signals and chassis ground. In addition, ID chassis ground shall be made externally available through an ID connector, compatible with the ATE system ground cable.

3.3.2.17 Interface device data package. There shall be a data package for each ID and it shall consist of the following elements

3.3.2.17.1 Interface device drawings. The ID drawings shall be in accordance with the DOD-D-1000, categories D/E/H form 3. These drawings shall included, at a minimum, schematic, assembly drawings, parts list and interface wire data list. Reference designations for electrical and electronic parts and equipment shall be in accordance with ANSI-STD-Y32.16A, and graphic symbols for electrical and electronic diagrams shall be in accordance with ANSI-STD-Y32.2

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3.3.2.17.2 Parts list. Parts list shall not be limited to the contractor's part numbers but shall include commercial equivalent part numbers if the part is commercially available.

3.3.2.17.3 Non-standard or special parts. If non-standard or special parts and materials are used for ID construction, then appropriate documentation as well as source control drawings for these items shall be provided by the contractor so as to be in accordance with 3.3.2.1.

3.3.2.17.4 Interface device pin wire list. The ID package shall include a pin list in tabular form illustrating the connections to the ATE interface and the UUT.

3.3.3 Test program instruction. The test program instruction (TPI) shall provide the ATE operator with the information required to initiate, and to perform all tests, including hook-up, program start procedures, disconnect, and UUT disposition. The contents and format of the TPI shall be prepared in accordance with MIL-M-38784, Appendix A of MIL-STD-2077, and as defined herein. The TPI shall be provided as hard copy.

3.3.3.1 Automated preparation. When systems or techniques capable of automatically processing and plotting system interconnection and test diagrams are utilized, continuous line plotting shall be required for all diagrams and in the case of symbols involving curved lines or unusual graphics, the best possible approximations shall be used.

3.3.3.2 TPI general requirements. The TPI general requirements are as follows

a. The TPI shall contain caution notes relative to removal and replacement of specific SRAs or components which are susceptible to damage or degradation from electrical static discharge (ESD). The caution note shall contain, as a minimum a statement that SRAs containing metal-oxide-semiconductor (MOS), insulated transistor-transistor logic (TTL) microcircuits, junction field-effect transistors FET, etc., in individual, network on-board capacity or hybrid form shall not be removed with power applied.

b. TPI pages shall be numbered sequentially at the bottom center of each page.

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3.3.4 Supplementary data. The form and content of supplementary data shall be as specified in MIL-STD-2077. In addition, these data shall include an ID data package in accordance with 3.3.2.17. The TPS supplementary data shall also include automatic test program generator (ATPG) modeling data as well as associated stimulus/ response/fault data printouts and tape write (TRIT) tape. A test diagram description is provided in Appendix B.

3.3.5 Additional OTPS requirements. An OTPS, as defined in 3.1, is an entity which consists of the merging of one or more TPSs. Specific requirements of each of the corresponding elements of an OTPS are stated in the following subparagraphs

3.3.5.1 Test program (TP). Each of the TPs of an OTPS shall reside on the test program medium (TPM). The TPM shall be structured in such a manner that it shall be possible to identify the individual test programs residing therein, i.e., a table of contents. The TPM shall include both source and object programs of each TP.

3.3.5.2 Interface device. The ID requirements of each TPS of the OTPS shall be satisfied by the resulting OTPS ID. This ID shall be designed such that no degradation of stimulus/response characteristics associated with a given TPS are affected by ID circuitry associated with another TPS of the OTPS.

3.3.5.3 Operational test program instructions (OTPI). Each of the TPIs of a OTPS shall be combined into an OTPI which shall be a hard copy and shall form a single document. This document shall include a table of contents listing the TPIs and their associated TPSs. Suitable separators and multiple volumes shall be employed to facilitate the use of the document.

3.3.5.4 Supplementary data. The supplementary data for each TPS shall be combined into an overall supplementary data document for the OTPS in hard copy and shall form a single document. This document shall include a table of contents listing the data packages and their associated TPSs. For uses of common data, e.g. ID data package, it need only be included once. Suitable separators, and multiple volumes shall be employed to facilitate the use of the document.

3.3.5.5 OTPS identification. The operator shall have the following additional means of identifying an OTPS.

a. Visual examination of label on the TPM or its container which describes the contents.

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b. Provisions of a header in ATE readable form.

c. Identification of all elements of an ID shall be in accordance with MIL-N-18307 for visual identification.

3.3.6 Master test program set index (MTPSI). A master test program set index shall be prepared in accordance with MIL-STD-2077.

3.3.7 Interchangeability. The test program shall function identically when run with any certified UUT, ID and ATE of the same configuration. Test stations containing the software and test modules required to execute a test program shall be considered to be the same configurations.

3.3.8 Safety. A system safety program in accordance with the requirements of MIL-STD-882 shall apply.

3.3.8.1 Personnel safety. Personnel safety consideration shall be in accordance with the requirements of "Personal hazard protection" of MIL-T-28800.

3.3.8.1.1 IDs. IDs shall be so constructed to present minimal hazards in themselves and protect users from UUT hazards. Where appropriate, the ID shall have warning legends affixed and shall enclose or shield hazardous UUTs.

3.3.8.1.2 TPIs. TPIs shall provide detailed descriptions of TPS and UUT hazards encountered when testing. In addition, operator warnings concerning special or unique dangers which may be encountered as a result of handling or analysis of the TPS or UUT in a manner contrary to normal testing shall be emphasized.

3.3.8.1.3 Test programs. Test programs shall warn operators of hazards by displaying appropriate messages on the ATE CRT. A warning message shall precede application of power and individual messages shall precede all test processes where an increased hazard is present. The TP shall, further, minimize operator contact with dangerous UUTs. In particular, probing should be effected by affixing clip probes with UUT power removed, and then restoring power and continuing test when the operator is clear.

3.3.8.2 Equipment safety. Provisions shall be included in the OTPS to protect the ATE as well as the UUT from physical and electrical hazards. The TPS

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shall be designed so that a failure of one portion of the TPS shall not cause damage to other portions of the TPS, ATE, or the UUT. Provisions shall also be included to prevent damage to the ATE due to a failure in the UUT.

3.4 Documentation. The OTPS documentation shall contain all of the documentation of the individual TPSs and associated UUT source data within the OTPS and shall be in accordance with the requirements of "Drawings and Documentation" of MIL-STD-2077. There is no requirement that all of the OTPS documentation be condensed into a single volume. It should be prepared in a manner however, such that any part of its contents can be easily referenced.

4. QUALITY ASSURANCE

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the general acceptance test procedure (GATP).

4.3 Quality conformance inspection. TPS and OTPS acceptance shall require the successful completion of the following, as described in the general acceptance test procedure (GATP)

a. TPS Design Verificaton Demonstration

b. Preproduction OTPS (First Article) Acceptance Demonstration

c. Initial Production OTPS Acceptance Demonstration

d. Production OTPS Acceptance Demonstration

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4.4 Methods of inspection. The OTPS shall be inspected and accepted in accordance with the Navy supplied general acceptance test procedure which shall be utilized to demonstrate compliance with this specification. The acceptance process shall use the ATE test station and the UUT configuration for which the OTPS was designed. All requirements contained in 3. of this specification shall be inspected by one or more of the following methods.

a. Visual - Visual and dimensional checks.

b. Analysis - Review of previous test data and theoretical analysis.

c. Test - Fit and functional test which requires the OTPS to be mated with its associated UUT and ATE.

4.4.1 Inspection (visual). The following requirements shall be verified by visual inspection when demonstrating the applicable section of the GATP.

Requirement	Requirement paragraph	Paragraph 4.3 applicability
Interface device (ID) definition	(3.1.2)	(b,c,d)
Panel ID	(3.1.2.1)	(b,c,d)
Cable set	(3.1.2.2)	(b,c,d)
Test fixture	(3.1.2.3)	(b,c,d)
Test program instruction (TPI) definition	(3.1.3)	(a,b,c,d)
Supplementary data definition	(3.1.4)	(a,b,c,d)
Physical characteristics	(3.2.2)	(b,c,d)
Interface device	(3.2.2.1)	(b,c)
Electrical	(3.2.2.1.1)	(b,c)
ID complexity	(3.2.2.1.3)	(b,c)

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Requirement	Requirement paragraph	Paragraph 4.3 applicability
ID optimization	(3.2.2.1.4)	(b,c)
ID expansion	(3.2.2.1.5)	(b,c)
ID size and weight	(3.2.2.1.6)	(b,c,d)
Signature testing	(3.2.2.1.7)	(b,c)
ID accessibility	(3.2.2.1.8)	(b,c)
ID enclosure	(3.2.2.1.9)	(b,c,d)
Drip-proof	(3.2.2.1.9.1)	(b,c,d)
Feet	(3.2.2.1.9.2)	(b,c,d)
Handles	(3.2.2.1.9.3)	(b,c,d)
External connectors	(3.2.2.1.9.4)	(b,c,d)
Attachment of nameplate	(3.2.2.1.9.5)	(c,d)
Chassis fastening	(3.2.2.1.9.6)	(b,c,d)
Test program maintainability	(3.2.4.1)	(a,b,c)
Test program language	(3.3.1.1)	(a,b)
Computer program definition requirement	(3.3.1.1.1)	(a,b)
Test program sequence	(3.3.1.3)	(a,b)
Identity tests	(3.3.1.3.1)	(a)
ATE survey test	(3.3.1.3.2)	(a)
Safe-to-turn-on tests	(3.3.1.3.3)	(a)
Power application	(3.3.1.3.4)	(a)
ID Confidence test	(3.3.1.3.5)	(a)
Performance test	(3.3.1.3.6)	(a)
Diagnostic tests	(3.3.1.3.7)	(a)
Adjustment/alignment tests	(3.3.1.3.8)	(a,b,c)
Removal of UUT power and stimuli	(3.3.1.3.9)	(a)

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Requirement	Requirement paragraph	Paragraph 4.3 applicability
ATE resource	(3.3.1.4)	(a,b)
Test cycling	(3.3.1.5)	(a,b)
Tolerance coning	(3.3.1.7)	(a,b,c)
Accuracy requirement	(3.3.1.7.1)	(a,b,c)
Automatic test program generator (ATPG)	(3.3.1.8)	(a,b,c)
Test program elements	(3.3.1.9)	(a)
Program partition	(3.3.1.9.1)	(a)
Entry points	(3.3.1.9.1.1)	(a)
Initial entry point	(3.3.1.9.1.1.1)	(a)
Performance entry points	(3.3.1.9.1.1.2)	(a)
Test group	(3.3.1.9.2)	(a)
Performance test groups	(3.3.1.9.2.1)	(a)
Test execution integrity check	(3.3.1.9.2.1.1)	(a)
Diagnostic test group	(3.3.1.9.2.2)	(a)
Test	(3.3.1.9.3)	(a)
Program heading and identification	(3.3.1.10)	(a)
Structured programming	(3.3.1.11)	(a)
Test program coding format	(3.3.1.12)	(a)
Comment usage	(3.3.1.12.2)	(a)
Test group and procedure delineator	(3.3.1.12.2.3)	(a)
Test body comments	(3.3.1.12.2.4)	(a)
Test numbering	(3.3.1.12.3)	(a)
Messages	(3.3.1.13)	(a)

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Requirement	Requirement paragraph	Paragraph 4.3 applicability
Standard messages	(3.3.1.13.1)	(a)
Non-standard messages	(3.3.1.13.2)	(a)
Self containedness	(3.3.1.15)	(a,b)
Test operator interface	(3.3.1.16)	(a)
ID design and construction	(3.3.2)	(b,c,d)
Parts and materials	(3.3.2.1)	(b,c)
Design stresses	(3.3.2.2)	(b,c,d)
Cables	(3.3.2.3)	(b,c,d)
Strain relief	(3.3.2.3.1)	(b,c)
Coding	(3.3.2.3.2)	(b,c,d)
Cable assembly configuration	(3.3.2.3.3)	(b,c,d)
Electrical connectors	(3.3.2.4)	(b,c,d)
Contacts	(3.3.2.4.1)	(b,c,d)
Spacing and quantity	(3.3.2.4.2)	(b,c,d)
Protective caps	(3.3.2.4.3)	(b,c,d)
Allowable internal ID connector types	(3.3.2.4.4)	(b,c,d)
Connector keying	(3.3.2.4.5)	(b,c,d)
Electromagnetic interference control (EMIC)	(3.3.2.5)	(b,c)
Maintenance of electronic assemblies	(3.3.2.6)	(b,c,d)
Printed circuit cards	(3.3.2.6.1)	(b,c,d)
Active devices	(3.3.2.6.2)	(b,c,d)
Standard buffering devices	(3.3.2.7)	(b,c,d)

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Requirement	Requirement paragraph	Paragraph 4.3 applicability
Terminal blocks	(3.3.2.8)	(b,c,d)
Interchangeability	(3.3.2.9) and (3.3.7)	(d)
Finishes	(3.3.2.11)	(b,c,d)
Protective finish	(3.3.2.11.1)	(b,c,d)
Color	(3.3.2.11.2)	(b,c,d)
Lacquer	(3.3.2.11.3)	(b,c,d)
Nameplates and product markings	(3.3.2.12)	(c,d)
Marking of nontype designated units, parts and assemblies	(3.3.2.12.1)	(b,c,d)
Identification of type designated units	(3.3.2.12.2)	(b,c,d)
Identification of type designated sets and groups	(3.3.2.12.3)	(b,c,d)
Workmanship	(3.3.2.13)	(a,b,c,d)
Safety	(3.3.2.14) and (3.3.8)	(a,b,c,d)
Human engineering/ human performance	(3.3.2.15)	(b,c,d)
Grounding	(3.3.2.16)	(b,c,d)
Interface device data package	(3.3.2.17)	(b,c,d)
Interface device drawings	(3.3.2.17.1)	(b,c,d)
Parts list	(3.3.2.17.2)	(b,c,)
Non-standard or special parts	(3.3.2.17.3)	(b,c)

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Requirement	Requirement paragraph	Paragraph 4.3 applicability
Interface device pin wire list	(3.3.2.17.4)	(b,c)
Test program instruction	(3.3.3)	(a,b,c)
Automated preparation	(3.3.3.1)	(a)
TPI general requirements	(3.3.3.2)	(a,b,c)
Supplementary data	(3.3.4)	(a,b,c)
Additional OTPS requirements	(3.3.5)	(b,c)
Interface device	(3.3.5.2)	(b,c)
Master test program set index (MTPSI)	(3.3.6)	(a)
Personnel safety	(3.3.8.1)	(a,b,c,d)
IDs	(3.3.8.1.1)	(a,b,c,d)
TPIs	(3.3.8.1.2)	(a,b,c)
Equipment safety	(3.3.8.2)	(a,b,c,d)
Documentation	(3.4)	(a,b,c)

4.4.2 Inspection (analysis). The following requirements shall be inspected by analysis when demonstrating the applicable section of the GATP.

Requirement	Requirement paragraph	Paragraph 4.3 applicability
Performance characteristics	(3.2.1)	(a,b,c)
Performance tests (PT)	(3.2.1.1)	(a,b,c)
UUT performance testing interface	(3.2.1.1.1)	(a,b,c,d)
Fault detection	(3.2.1.2)	(a,b,c)

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Requirement	Requirement paragraph	Paragraph 4.3 applicability
Diagnostic fault isolation tests	(3.2.1.3)	(a,b,c)
UUT fault isolation testing interface	(3.2.1.3.1)	(a,b,c)
Ambiguity groups	(3.2.1.3.2)	(a)
WRA fault isolation	(3.2.1.3.2.1)	(a)
SRA fault isolation	(3.2.1.3.2.2)	(a)
Test point interface	(3.2.2.1.2)	(a)
ID complexity	(3.2.2.1.3)	(b,c)
Test set-up time	(3.2.2.1.10)	(b,c)
Useful life	(3.2.2.2)	(a,b)
Operational service life	(3.2.2.2.1)	(a,b)
Storage life	(3.2.2.2.2)	(a,b)
Reliability	(3.2.3)	(a,b)
ID mean-time-between-failure (MBTF)	(3.2.3.1)	(b)
Maintainability	(3.2.4)	(a,b)
ID maintainability	(3.2.4.2)	(b)
ID self-test	(3.2.4.2.1)	(b,c)
ID mean time to repair (MTTR)	(3.2.4.2.2)	(b,c)
Service conditions (environmental)	(3.2.5)	(b,c)
Temperature-altitude combination	(3.2.5.1)	(b)
Humidity	(3.2.5.2)	(b)
Vibration	(3.2.5.3)	(b)

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Requirement	Requirement paragraph	Paragraph 4.3 applicability
Transportability	(3.2.6)	(b)
Required UUT documentation	(3.3.1.2)	(a)
Single/multiple fault criteria	(3.3.1.6)	(a)
Test efficiency	(3.3.1.14)	(a)
Test program instruction	(3.3.3)	(a,b,c)
Interface device	(3.3.5.2)	(b,c,d)

4.4.3 Inspection (test). The following requirements shall be inspected by testing when demonstrating the applicable section of the GATP, at a Navy designated site after completion of TPS integration

Requirement	Requirement paragraph	Paragraph 4.3 applicability
Interface device (ID) definition	(3.1.2)	(b,c,d)
Performance characteristics	(3.2.1)	(a,b,c)
Performance tests (PT)	(3.2.1.1) and (3.3.1.3.6)	(a)
UUT performance testing interface	(3.2.1.1.1)	(a,b,c,d)
Fault detection	(3.2.1.2)	(a,b,c)
Diagnostic fault isolation tests	(3.2.1.3) and (3.3.1.3.7)	(a)
WRA fault isolation	(3.2.1.3.2.1)	(a)
Adjustment/alignment routine	(3.2.1.4) and (3.3.1.3.8)	(a)
ATE survey test	(3.2.1.5) and (3.3.1.3.2)	(a,b,c)

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Requirement	Requirement paragraph	Paragraph 4.3 applicability
Identity (signature) tests	(3.2.1.6) and (3.3.1.3.1)	(a)
Safe-to-turn-on-test	(3.2.1.7) and (3.3.1.3.3)	(a,b,c)
ID confidence test	(3.2.1.8) and (3.3.1.3.5)	(b,c)
Interface device (ID)	(3.2.2.1)	(b,c)
Electrical	(3.2.2.1.1)	(b,c)
Test set-up time	(3.2.2.1.10)	(b,c)
ID self-test	(3.2.4.2.1)	(b,c)
Test program	(3.3.1)	(a)
Test program sequence	(3.3.1.3)	(a)
Power application	(3.3.1.3.4)	(a)
Removal of UUT power and stimuli	(3.3.1.3.9)	(a)
Test cycling	(3.3.1.5)	(a)
Tolerance coning	(3.3.1.7)	(a,b,c)
Test program elements	(3.3.1.9)	(a)
Program partition	(3.3.1.9.1)	(a)
Entry points	(3.3.1.9.1.1)	(a)
Initial entry point	(3.3.1.9.1.1.1)	(a)
Performance entry point	(3.3.1.9.1.1.2)	(a)
Performance test groups	(3.3.1.9.2.1)	(a)

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Requirement	Requirement paragraph	Paragraph 4.3 applicability
Test execution integrity check	(3.3.1.9.2.1.1)	(a)
Diagnostic test group	(3.3.1.9.2.2)	(a)
Test	(3.3.1.9.3)	(a)
Statement	(3.3.1.9.4)	(a)
Test numbering	(3.3.1.12.3)	(a)
Messages	(3.3.1.13)	(a)
Test operator interface	(3.3.1.16)	(a)
Test program instruction	(3.3.3)	(a,b,c)
Interchangeability	(3.3.7)	(d)
Test programs	(3.3.8.1.3)	(a)

4.5 TPS integration log. An TPS integration log shall be maintained to provide a history of TPS integration and test activities. The following types of information shall be entered in the log

a. Performance tests debug results.

b. Diagnostic tests debug results.

c. Results of all fault insertions during integration activities.

d. Description of all test program changes implemented during integration activities.

e. ID component failures and corrective maintenance or design changes implemented.

f. ATE test station failures and corrective maintenance performed.

g. Serial numbers of UUTs and ATE test stations utilized.

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5. PACKAGING

5.1 Interface device.

5.1.1 Preservation and packaging. The ID shall be preserved and packaged in accordance with MIL-P-116.

5.1.2 Packing. Shipments shall be packed in exterior type shipping containers in a manner that will ensure safe transportation at the lowest rate to the point of delivery. Containers shall meet the requirements of the consolidated freight classification rules and regulations of the common carriers as applicable to the mode of transportation.

5.1.3 Marking. All packages and containers of the packaged equipment shall be marked to suitably identify the contents in accordance with the requirements of MIL-STD-129, and shall include any special markings as required by carrier rules and regulations. In addition, all packages and containers shall be marked to include the contract number and total number of pieces. Items that require special treatment or that have undergone special cleaning, etc., shall be so marked on the exterior and interior of the packages and containers.

5.2 Test Program Medium (TPM).

5.2.1 Magnetic tape reels, discs, and punched cards.

5.2.1.1 Magnetic tape reels and disc. Magnetic tape reels and discs shall be packed in materials that protect the recorded information from accidental degaussing by stray fields. The shipping containers shall be moisture and dust proof. The reels and discs shall be packed so that no physical distortion occurs in handling.

5.2.1.2 Punched cards. Punched cards, if applicable, shall be packed in moisture and dust proof containers. The cards shall be packed so that no physical deformation occurs in handling.

5.3 Operational test program instruction (OTPI), supplementary data and OTPS associated documentation. The OTPI (TPI) supplementary data and OTPS associated documentation shall be packed in moisture and dust proof containers. They shall be packed so that no physical deformation occurs in handling.

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6. NOTES

6.1 Definitions. Definitions shall be those provided in 6.3 of MIL-T-28800 and the following

6.1.1 Contractor. When this document or any other associated document refers to the contractor, it shall be interpreted to mean the seller.

6.1.2 End-to-end run time. End-to-end run time is the program execution time required to determine that a good UUT is RFI.

6.1.3 Government. United States of America Government.

6.1.4 Military. United States of America Military.

6.1.5 Operating service life. The operating service life is the total accumulative time that the unit is actually operating or in an "on-line" status with voltage or mechanical fluid sources applied. The time may be in terms of hours, cycles, or other related parameters.

6.1.6 Procuring activity. When this document or any other associated document refers to the procuring activity, and like terms, it shall be interpreted to mean U.S. Navy.

6.1.7 Shop replaceable assembly (SRA). An SRA is a generic term which includes all the packages within a WRA. Each SRA shall be single plug-in assembly. All parts, except the chassis, chassis wiring, and chassis mounted component as specified herein shall be mounted on the SRAs. No SRA shall be hard wired to the chassis or other SRA wiring.

6.1.8 Signature/configuration resistor. Signature/configuration resistors are unique value resistors located in/on each ID and are used for automatic verification that the correct OTPS is being used with the UUT.

6.1.9 Storage life. The storage life is the total accumulative time that the equipment may be kept in storage before or after being used and still have the specified operational service life without component or part replacement, adjustment or unscheduled maintenance action.

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6.1.10 Test accuracy ratio (TAR). Test accuracy ratio (TAR) is defined as the ratio of the measurement uncertainty of the UUT to the measurement uncertainty of the ATE. For example, if it is required that a UUT output be accurate to five (5%) percent and the ATE accuracy measuring the parameter is one (1%) percent, then the TAR is five-to-one (5 1).

6.1.11 Useful life. The useful life is the period from the time of delivery of the equipment to the procuring agency until its identity is destroyed by classifying it as salvage or it is subject to cannibalization or both. The useful life includes the operating service life and the storage life.

6.1.12 Weapon replaceable assembly (WRA). An WRA is a generic term that refers to a major assembly of an equipment which is separately installed/replaced in a SE item or an aircraft. The term also refers to "units", "test drawers", and "IDs".

6.2 General.

6.2.1 Alternative requirements. As part of the contractor's design and development activities relative to the equipment requirements herein, the contractor is encouraged to suggest alterations to any of these specific requirements which in the contractor's opinion will result in a better overall system in consideration of performance, logistics, cost, delivery, and all other pertinent factors.

6.3 Acronyms.

ATE	Automatic Test Equipment
ATG	Automatic Test Generation
ATLAS	Abbreviated Test Language for All Systems
ATP	Acceptance Test Procedure
ATPG	Automatic Test Program Generator
BIT	Built-In-Test
CMOS	Complementary Metal Oxide Semiconductor
CRT	Cathode Ray Tube
EMI	Electromagnetic Interference
EMIC	Electromagnetic Interference and Compatibility
ESD	Electrical Static Discharge
HTS	Hybrid Test Station
IC	Integrated Circuit
ID	Interconnection Device

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MOS	Metal Oxide Semiconductor
MTBF	Mean Time Between Failures
MTBMA	Mean Time Between Maintenance Action
MTBUMA	Mean Time Between Unscheduld Maintenance
MTPSI	Master Test Program Set Index
MTTR	Mean-Time-To-Repair
OTPI	Operational Test Program Instruction
OTPS	Operational Test Program Set
PT	Performance Test
RFI	Ready For Issue
SRA	Shop Replaceable Assembly
STTO	Safe-To-Turn On
TAR	Test Accuracy Ratio
TD	Test Diagram
TP	Test Program
TPI	Test Program Instruction
TPS	Test Program Set
TPM	Test Program Medium
TTL	Terminator-Transistor Logic
UUT	Unit Under Test
WRA	Weapon Replaceable Assembly

Preparing Activity
NAVY - AS
(Project no. 6625-N-800)

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APPENDIX A

STANDARD BUFFERING DEVICES

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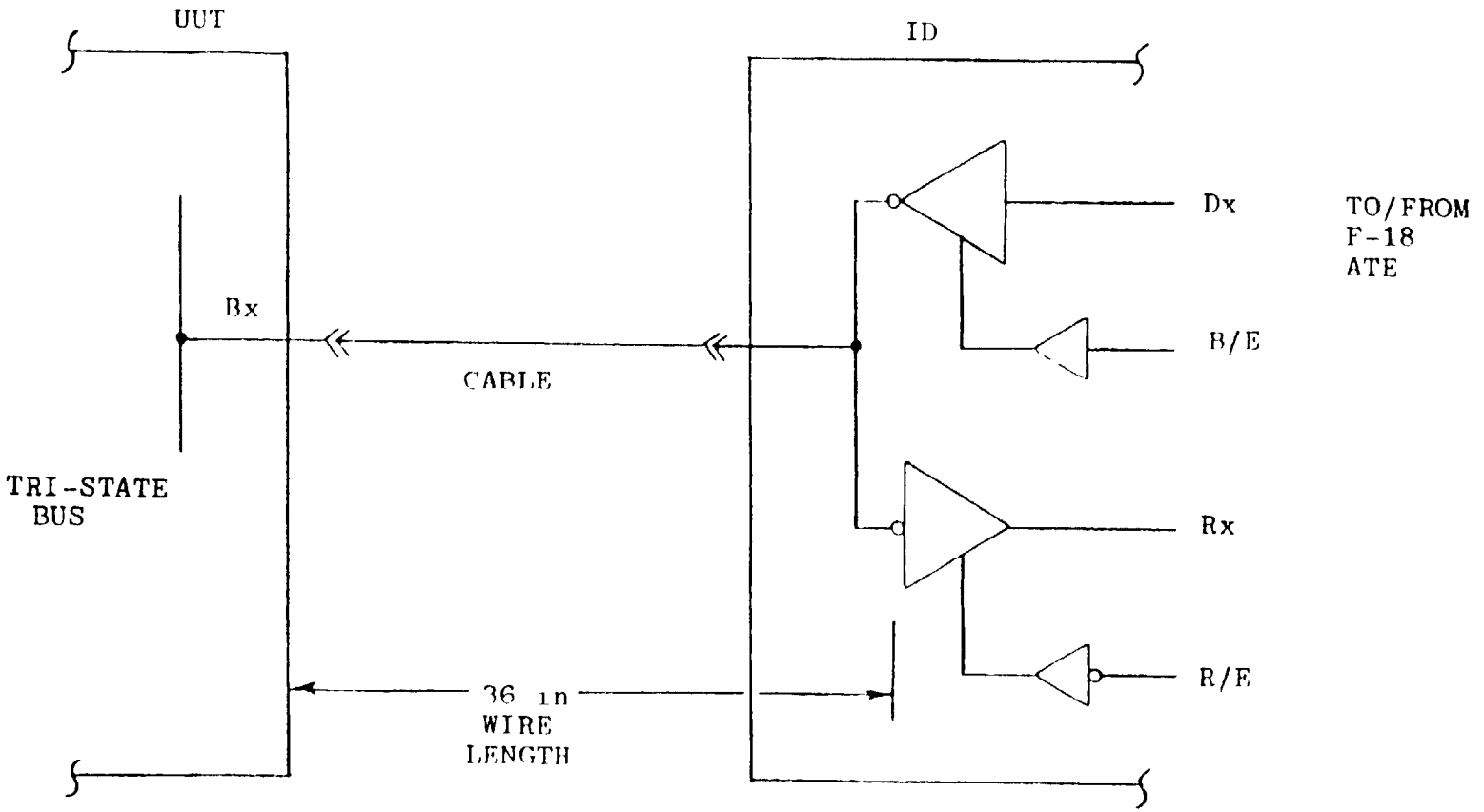
STANDARD BUFFERING DEVICES

DEVICE NO.	DESCRIPTION
8T26A, 54LS242	Tri-state bus transceiver
9614,55114	TTL dual differential line driver
9615,55115	TTL dual differential line receiver
54LS08	TTL low power Schottky AND gate
MM78C30	CMOS dual differential line driver
4041A	CMOS true/compliment buffer
54LS14	TTL single ended buffer
54LS32, 54128	TTL single ended line driver
LF156, LF256	Linear buffer
LH0063	Video buffer
AM687	ECL dual differential receiver
10524	Translator, TTL to ECL
10525	Translator, ECL to TTL

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TRI-STATE BUS TRANSCEIVER

54LS242, 8T26A



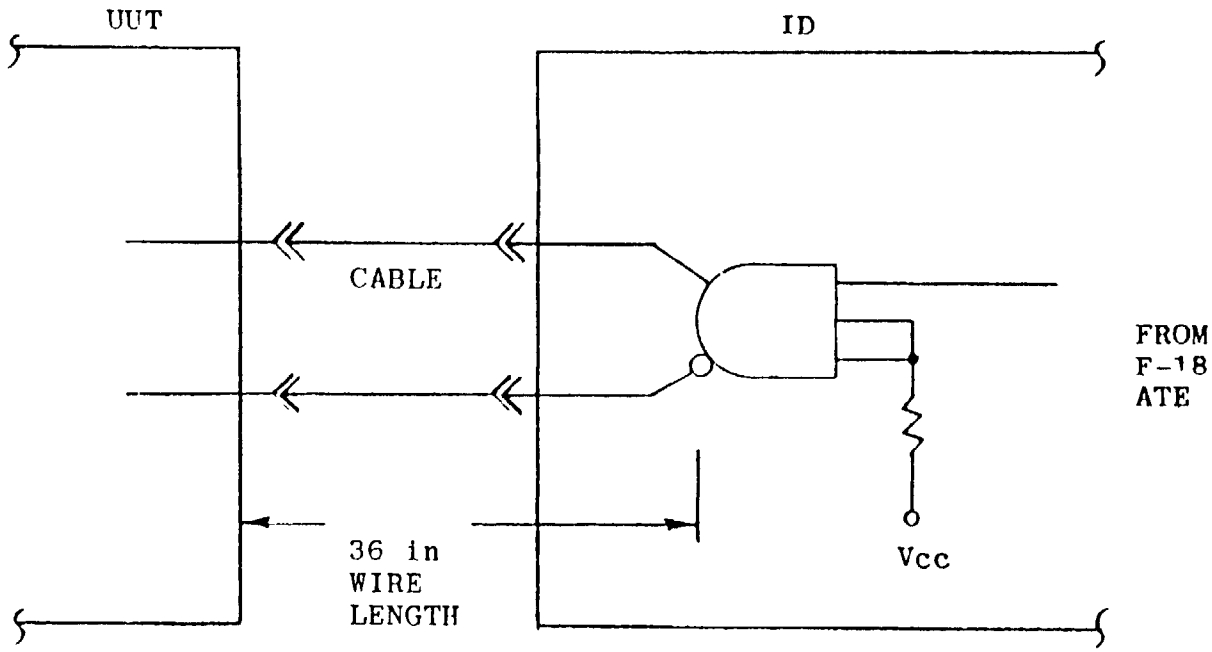
NOTE B/E IS BUS ENABLE } COMMON TO 4
R/E IS RECEIVER ENABLE } DRIVER/RECEIVER
Dx IS INPUT DRIVER TO BUS-x
Rx IS OUTPUT RECEIVER FROM BUS-x
Bx IS INPUT/OUTPUT BUS FOR
DRIVER/RECEIVER, x

REQUIREMENT TO BE USED WITH TRI-STATE BUSES

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TTL DUAL DIFFERENTIAL LINE DRIVER

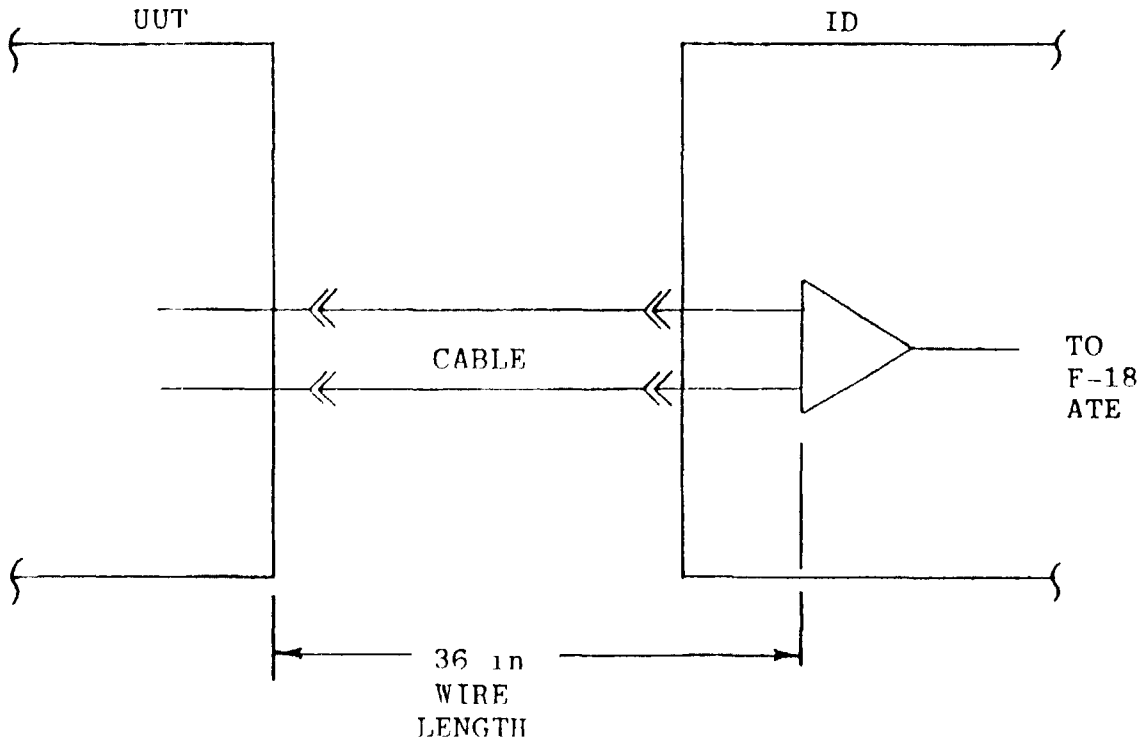
9614, 55114



REQUIREMENT TO BE USED WITH DIFFERENTIAL DEVICES WHICH CANNOT BE DRIVEN DIRECTLY FROM F-18 ATE (TTL LOGIC)

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TTL DUAL DIFFERENTIAL LINE RECEIVER
9615, 55115

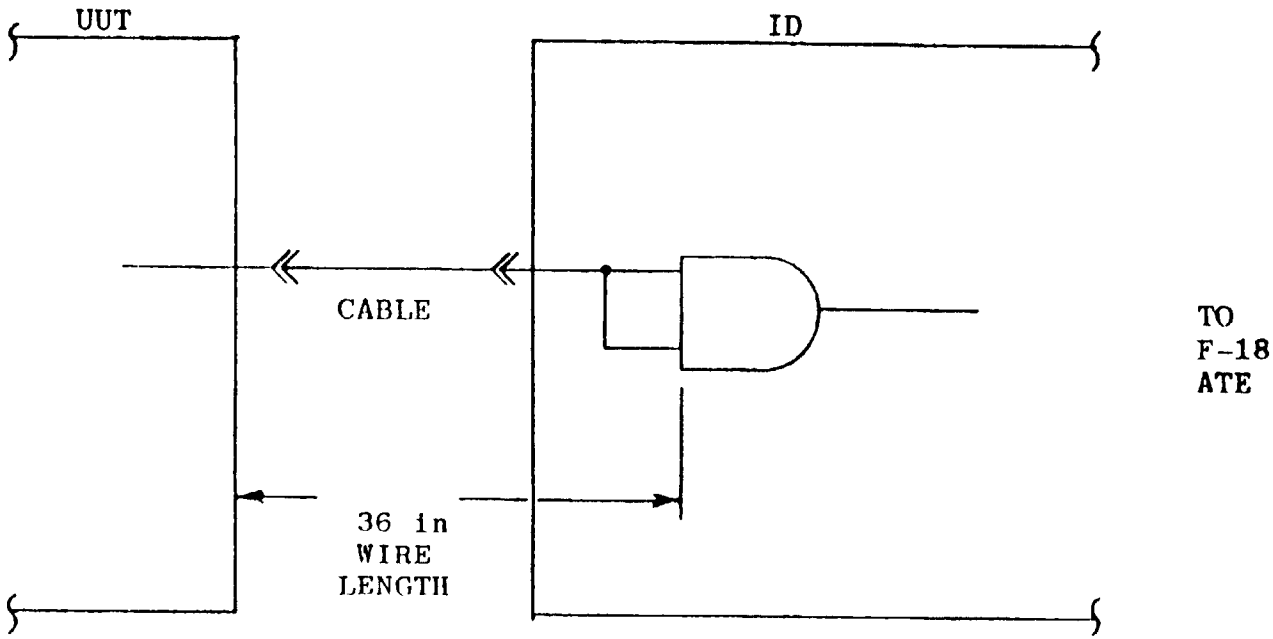


REQUIREMENT: TO BE USED WITH DIFFERENTIAL SIGNALS WHICH CANNOT DRIVE DIRECTLY INTO F-18 ATE.

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TTL LOW POWER SCHOTTKY "AND" GATE

54LS08

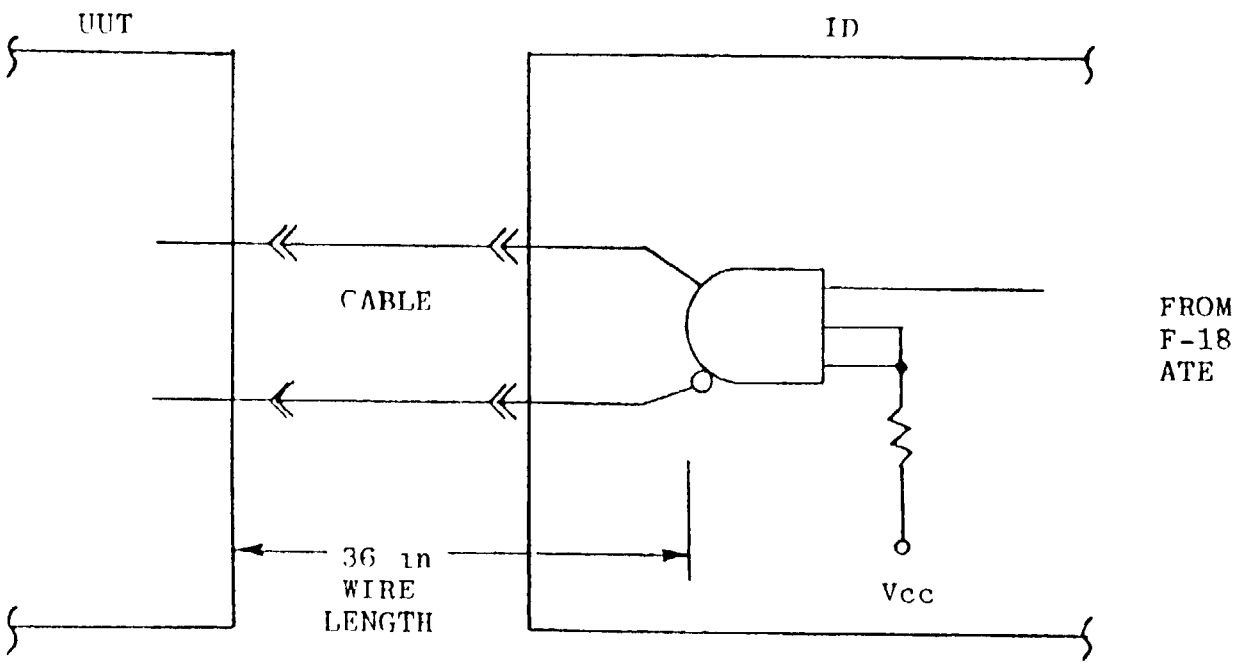


REQUIREMENT FOR USE WITH LOW POWER SCHOTTKY DEVICES WHICH CANNOT DRIVE DIRECTLY INTO F-18 ATE (TTL LOGIC) DUE TO INPUT CURRENT REQUIREMENTS

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CMOS DUAL DIFFERENTIAL LINE DRIVER

MM78C30

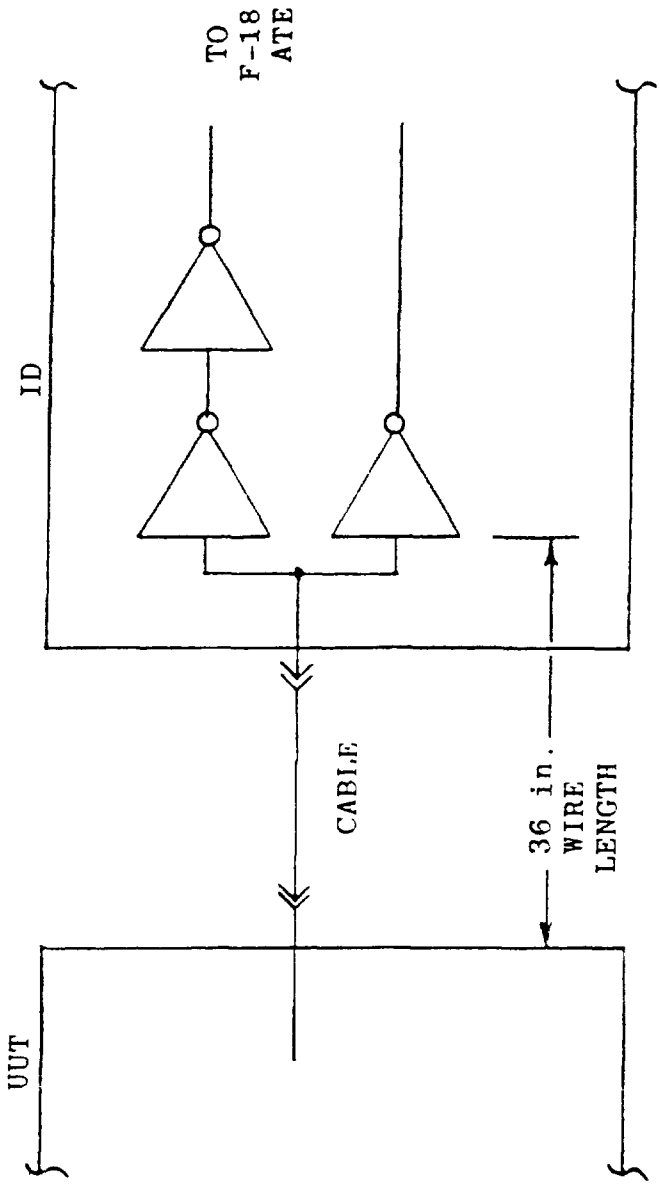


REQUIREMENT: FOR USE WITH CMOS DIFFERENTIAL DEVICES WHICH CANNOT BE DRIVEN DIRECTLY FROM F-18 ATE (TTL LOGIC).

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CMOS, TRUE/COMPLEMENT BUFFER

4041A

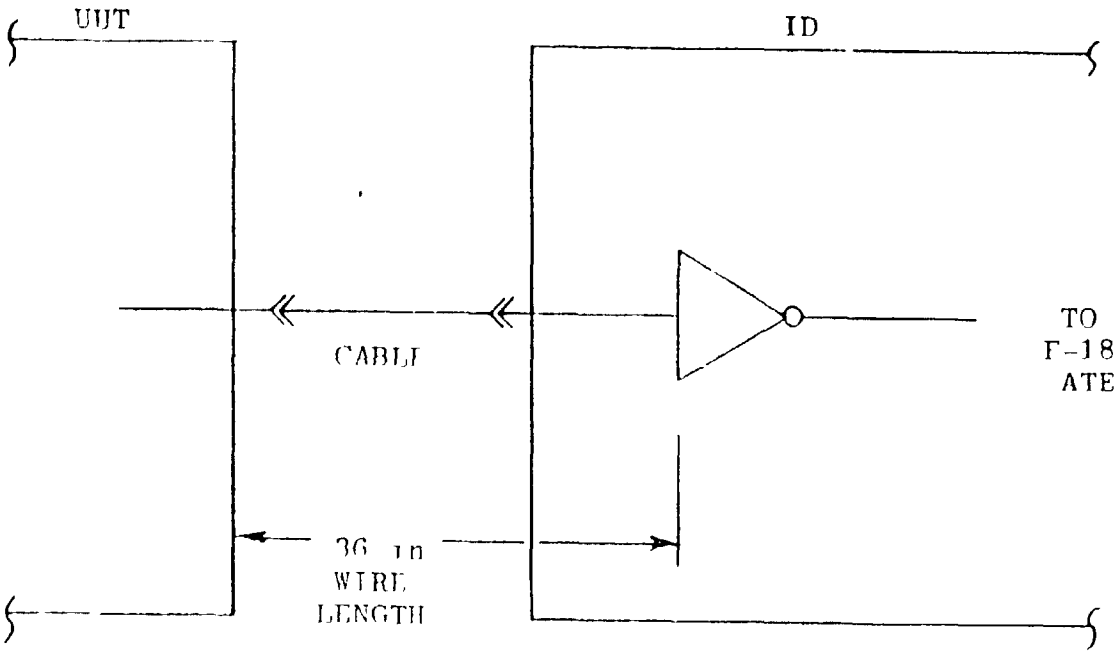


REQUIREMENT: FOR USE WITH CMOS DEVICES WHICH CANNOT DRIVE DIRECTLY INTO F-18 ATE (TTL LOGIC) DUE TO INPUT CURRENT REQUIREMENTS.

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TTL SINGLE-ENDED BUFFER

54LS14 (SCHMITT TRIGGER)

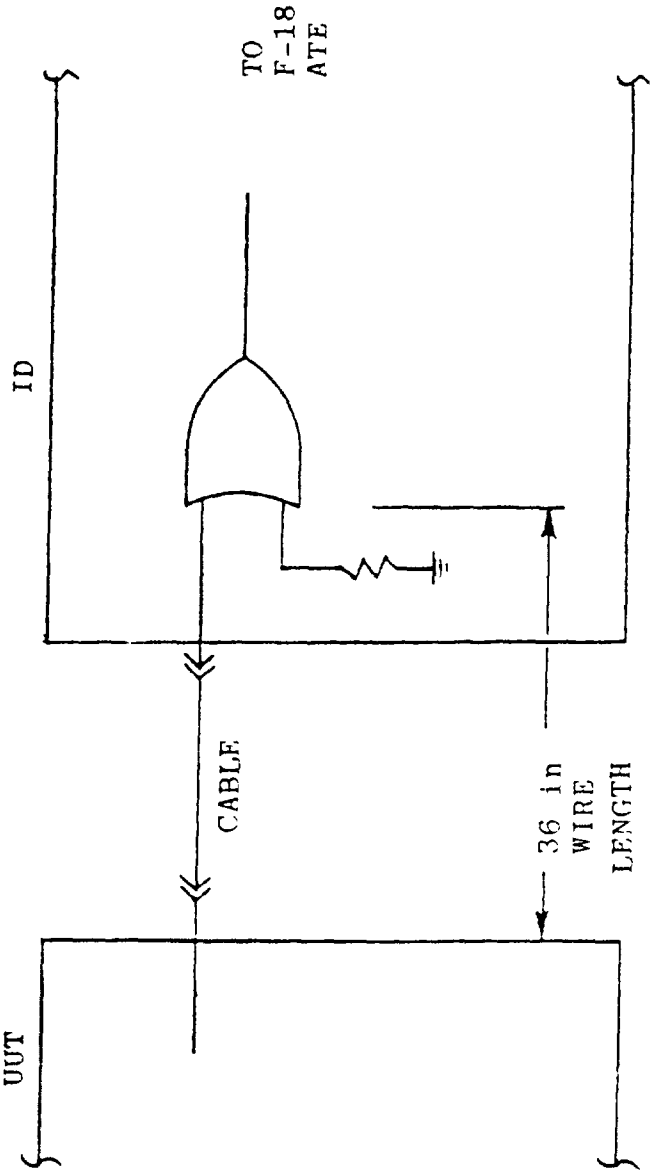


REQUIREMENT TO BE USED WITH SIGNALS WHICH REQUIRE A SHARPENING OF LEADING AND TRAILING EDGES

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TTL SINGLE-ENDED LINE DRIVER

54LS32

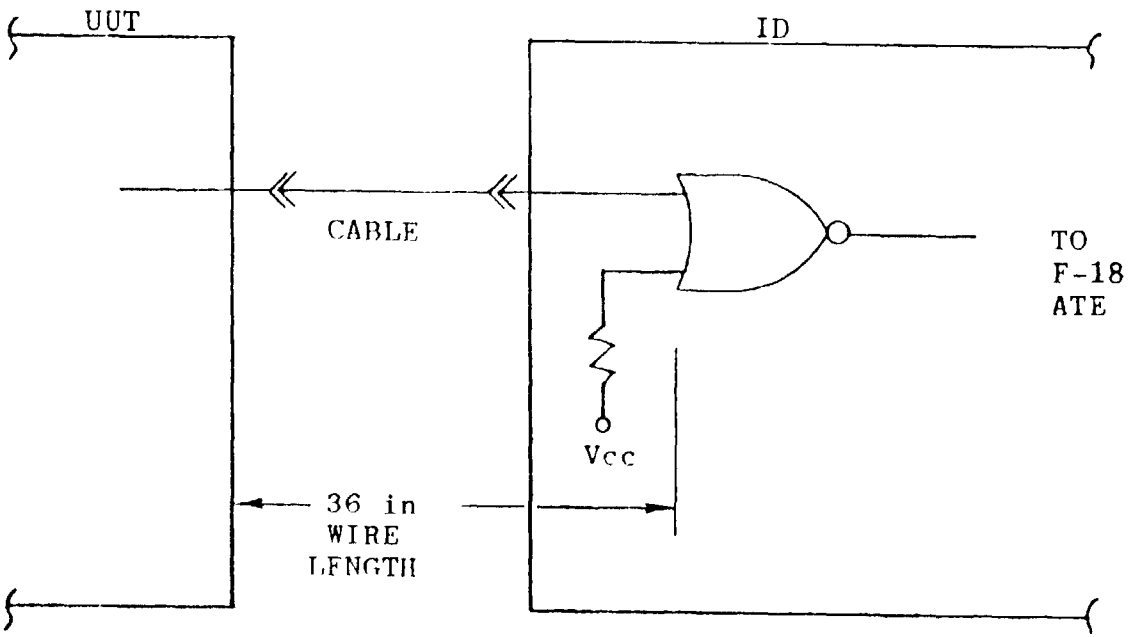


REQUIREMENT FOR USE WITH TTL DEVICES WHICH CANNOT DRIVE DIRECTLY INTO F-18 ATE (TTL LOGIC) DUE TO INPUT CURRENT REQUIREMENTS.

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TTL SINGLE-ENDED LINE DRIVER

54128

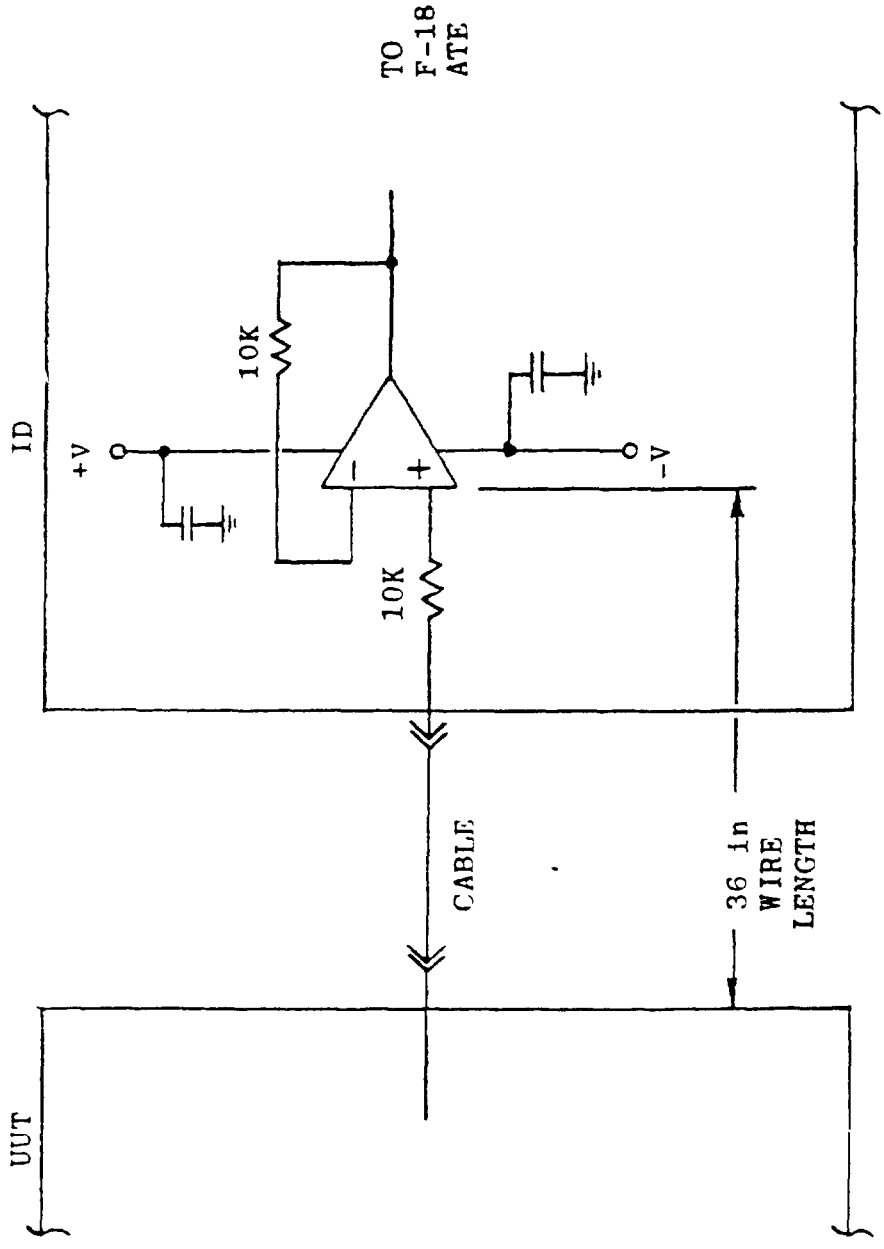


REQUIREMENT FOR USE WITH TTL DEVICES WHICH CANNOT DRIVE DIRECTLY INTO F-18 ATE (TTL LOGIC) DUE TO INPUT CURRENT REQUIREMENTS

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LINEAR BUFFER

LF156, LF256

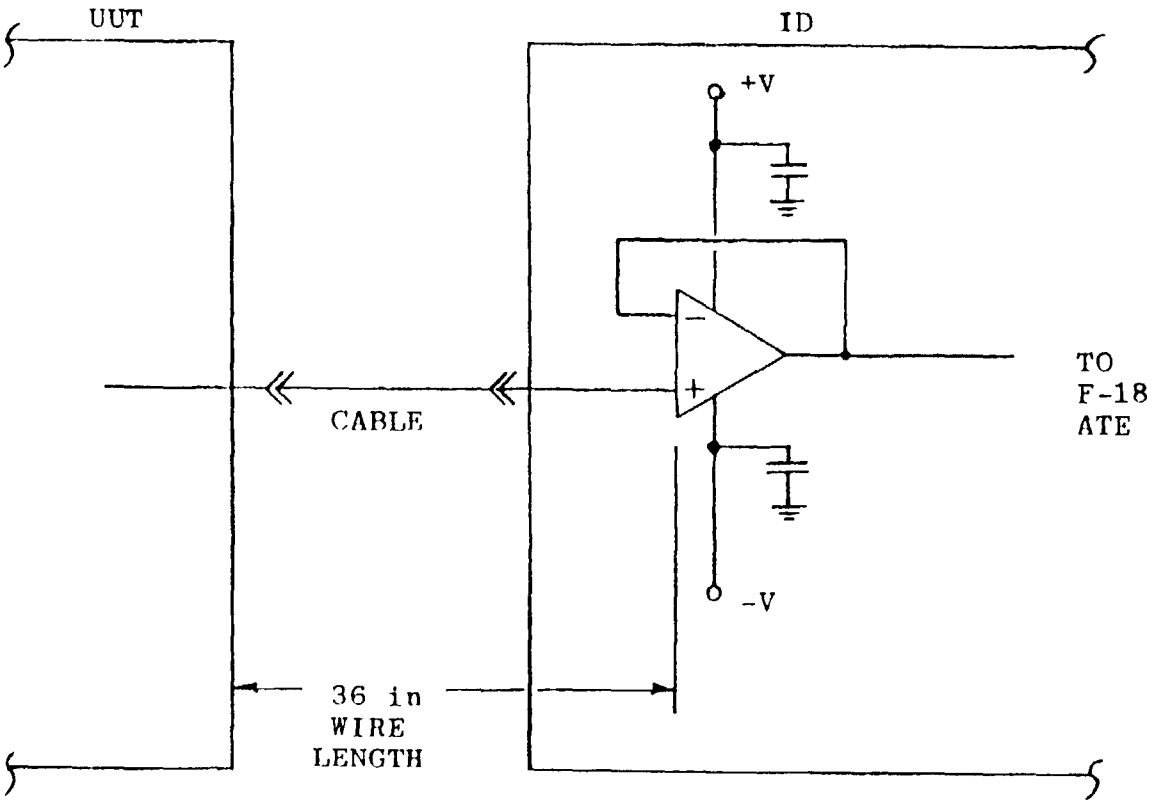


REQUIREMENT: TO BE USED WITH ANALOG SIGNALS WHICH CANNOT DRIVE DIRECTLY INTO F-18 ATE DUE TO SIGNAL LOSSES IN THE ID CABLE SET.

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VIDEO BUFFER

LH0063

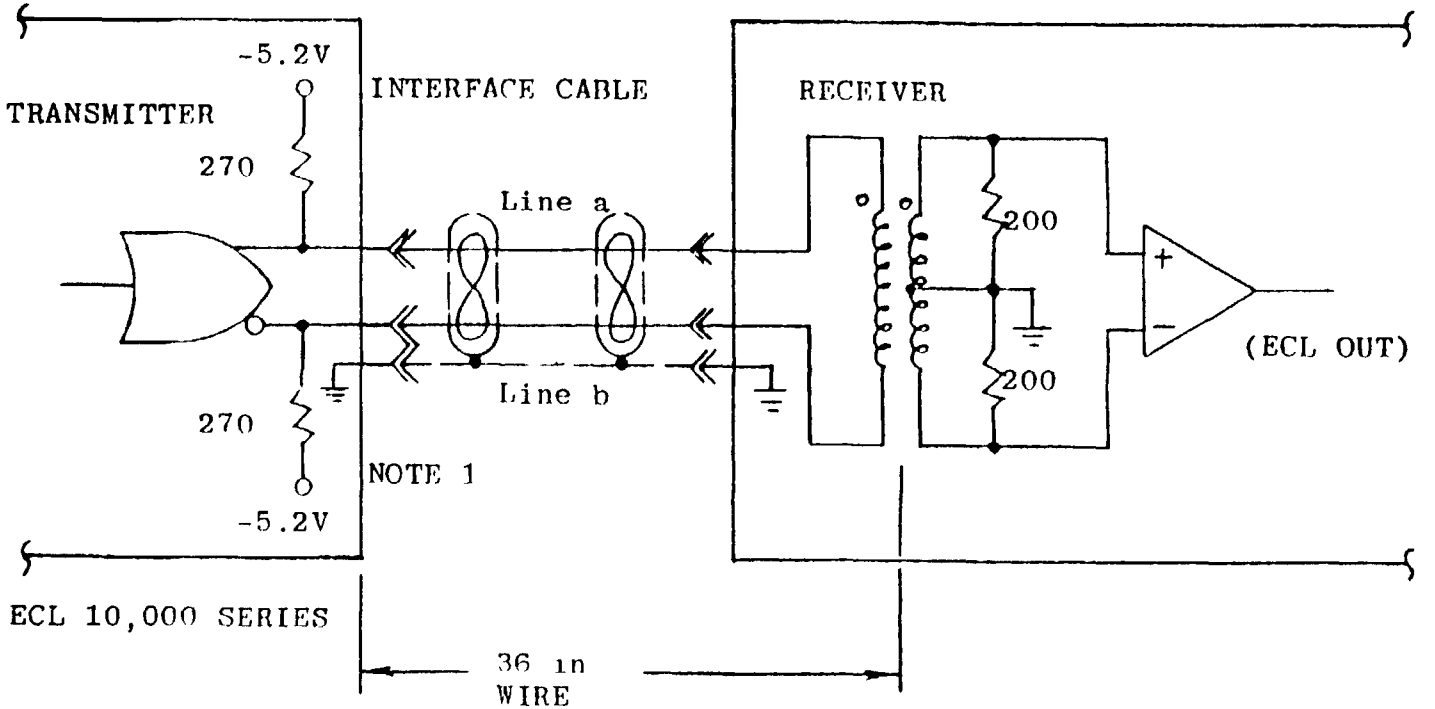


REQUIREMENT: TO BE USED WITH VIDEO SIGNALS WHICH CANNOT DRIVE DIRECTLY INTO F-18 ATE DUE TO SIGNAL LOSSES IN THE ID CABLE SET.

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(A) HIGH-SPEED ECL INTERFACE CIRCUIT-DUAL DIFFERENTIAL RECEIVER

AM687



LOGIC HIGH	LINE a	-0.6 to -1.0V
	b.	-1.4 to -1.9V
LOGIC LOW	LINE a.	-1.4 to -1 9V
	b	-0 6 to -1 0V

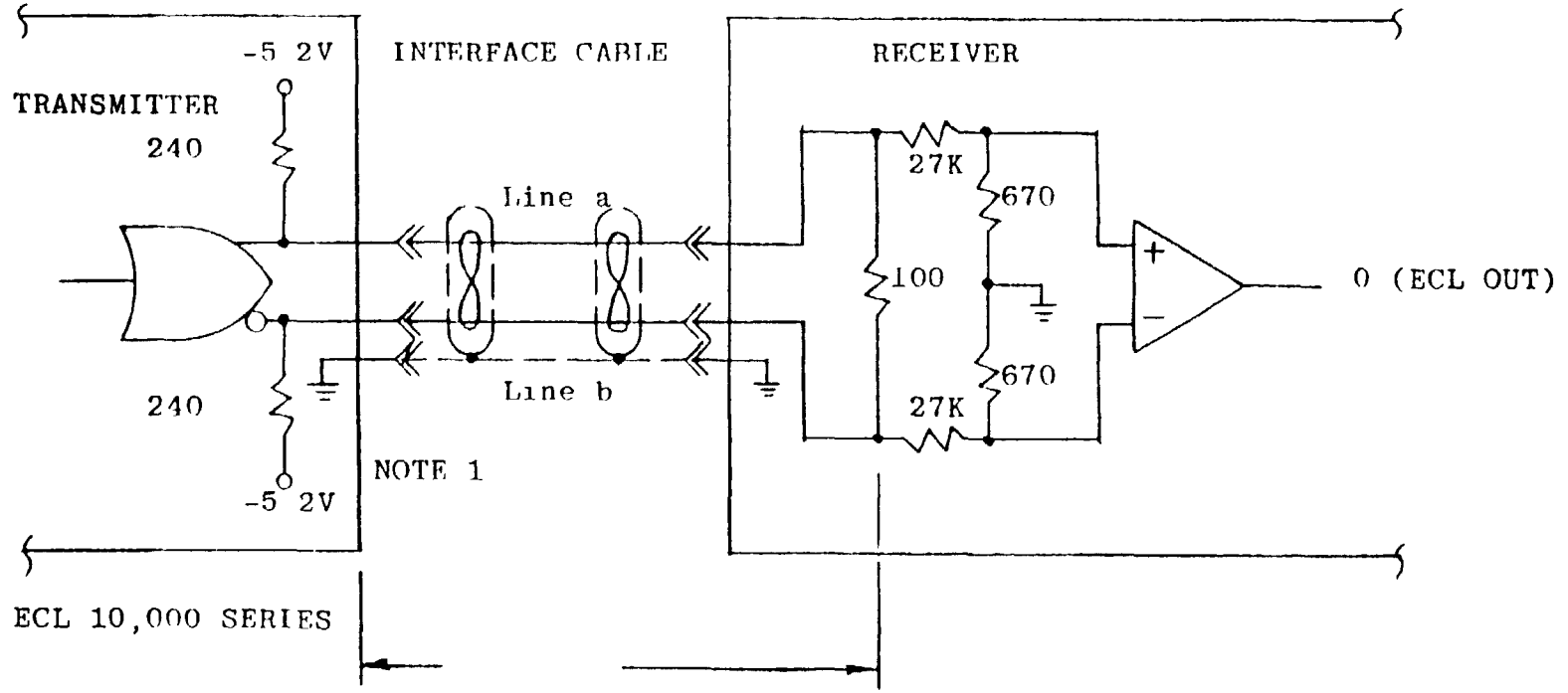
DATA RATE 50 to 70 MHz

NOTE 1. SHIELD SHALL BE GROUNDED TO CHASSIS IN TRANSMITTER AND RECEIVER UNITS WITH SHORTEST POSSIBLE LEADS

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(B) STANDARD ECL INTERFACE CIRCUIT-DUAL DIFFERENTIAL RECEIVER

AM687



DATA RATE = UP TO 50MHz

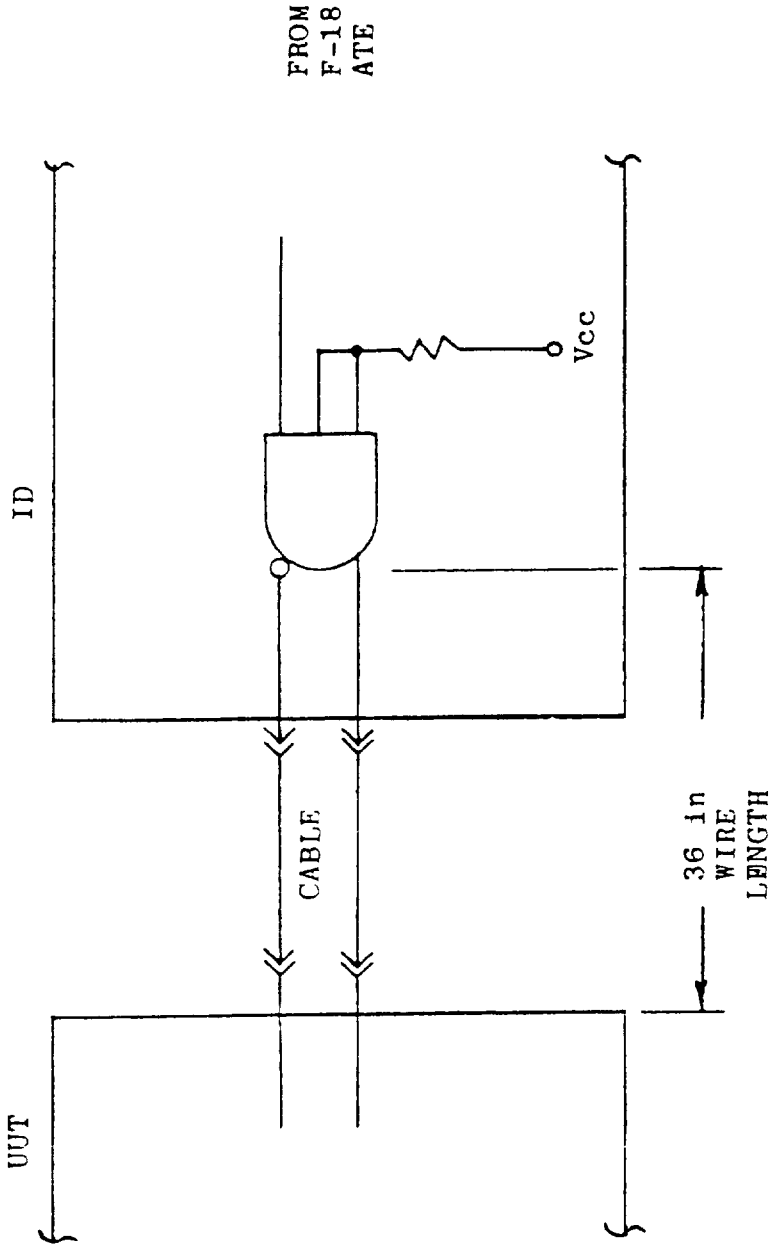
LOGIC HIGH·	LINE a ·	-0 6 to -1.0V
	b	-1.4 to -1.9V
LOGIC LOW·	LINE a ·	-1.4 to -1.9V
	b	-0.6 to -1 0V

NOTE 1· SHIELD SHALL BE GROUNDED TO CHASSIS IN TRANSMITTER AND RECEIVER UNITS WITH SHORTEST POSSIBLE LEADS

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TRANSLATOR, TTL TO ECL

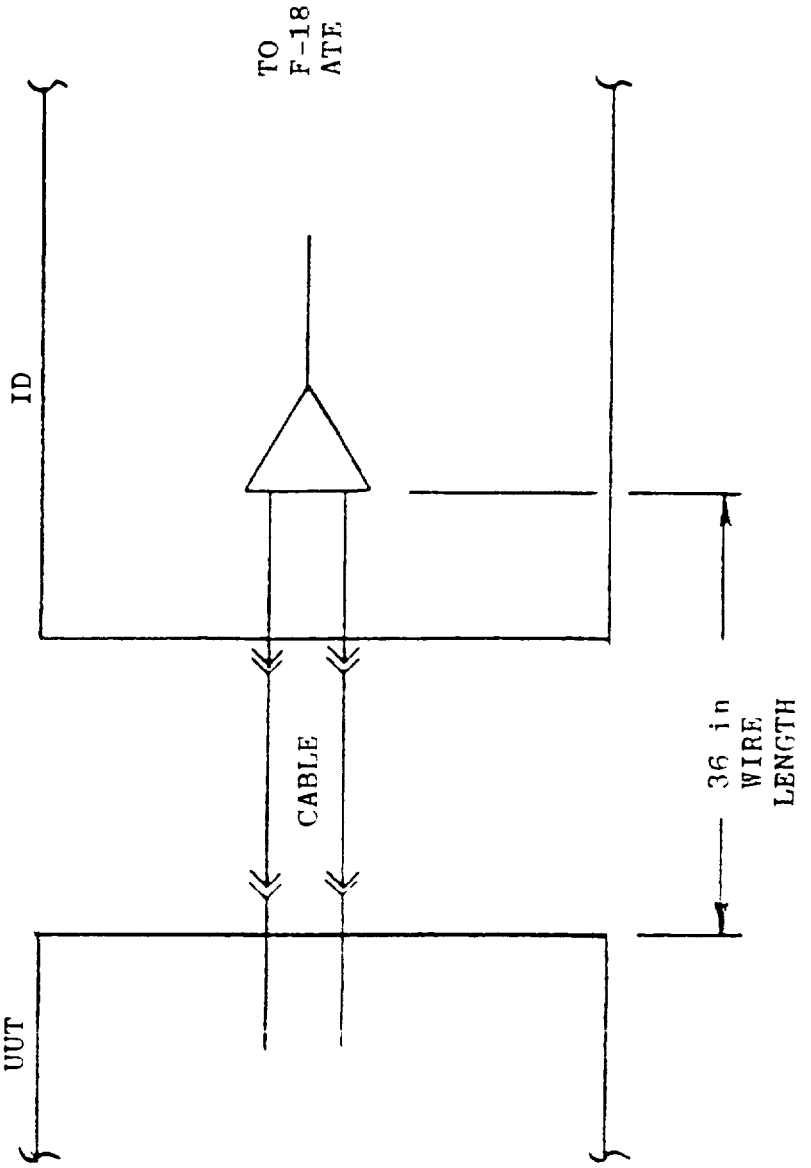
10524



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TRANSLATOR,ECL TO TTL

10525



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APPENDIX B

TEST DIAGRAM DESCRIPTION

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APPENDIX B

Test Diagram Description

10. SCOPE

10.1 Scope. This appendix describes the purpose and the requirements for generating test diagrams (TD).

20. REFERENCED DOCUMENTS

Not Applicable

30. REQUIREMENTS

30.1 General requirements. The test diagram (see figure 1) shall provide an explicit definition of the active electrical interface between the ATE, ID, and the UUT. The supplementary data shall contain a test diagram for each test or group wherein the basic system interconnect is maintained. Each electrical path shall be connector and connector pin identified through its entire interconnection, UUT ID'ATE. The signal path through the ID shall identify all internal SRA pin/jack connections as the signal proceeds through the ID. All simple circuits (i.e., resistors, diodes) shall be identified by reference designation. Where path variables within the ATE form a part of the test requirements, the applicable switching devices and alternate destinations shall be identified and functionally illustrated. Complex networks and circuitry shall be identified as functional blocks. The test diagram is intended to provide information relevant to the test being performed. Those portions of the interconnection which remain unchanged for a number of tests (power, ground, etc.) should not be repeated. Different tests which require minor modification to the electrical interface may be identified on the basic test diagram if the required change does not exceed ten (10% percent of the total field. Adequate notation defining the variation must be included in the test diagram. Where more than one test diagram, with or without annotated variations, is required with no basic changes to significant portions of the basic electrical interface, this unchanging portion may be identified by a separate test diagram with appropriate notations on all associated test diagrams. Related test diagrams may be shown on the same sheet. Where multiple entries are made, they shall be clearly separated

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and identified. Notes appropriate to the entire set will be included in a single note field (bottom left) with positive indication that they associate with all test diagrams. Notes peculiar to single test diagrams shall appear directly under the specific test diagram involved. For digital tests which require a large number of signal lines, a given test diagram may be used together with a test table(s) indicating the active signals on a group or per test basis. For those digital tests wherein the source language program listing annotation specifically defines the active signal lines, the table entry shall reference the source language program listing. A test diagram shall be identified with a test diagram number to facilitate reference in the text of the TPI, and an index of test numbers/statement numbers to test diagram numbers shall be provided.

30.1.1 Comments to figure 1.

a. UUT nomenclature and part number. (See balloon 1.)

b. Cable reference designator. (See balloon 2.)

c. Number indicates active path for identically numbered test(s) (see balloon 3), one of which will identify the test either on the line following the signal or in the notes. Direction of symbol pointer indicates signal flow. Bi-directional signal paths shall be indicated by a dual pointer (⟷).

d. ID SRA information (see balloon 4) to include

(1) SRA reference designator
(2) SRA pin/jack connections
(3) Component reference designator

e. UUT signal name. (See balloon 5.)

f. Dashed line used to enclose a complex ID function consisting of circuitry on several SRAs. (See balloon 6.)

g. ID chassis mounting point. (See balloon 7.)

h. Notes are used to reference common signals (power, ground) of the UUT and ID to the appropriate test diagram(s). Notes referencing a test(s) will be footnoted by the use of the pointer symbol and number, and pertain only to the test(s) identified. (See balloon 8.)

i. All connector designators shall be specified.

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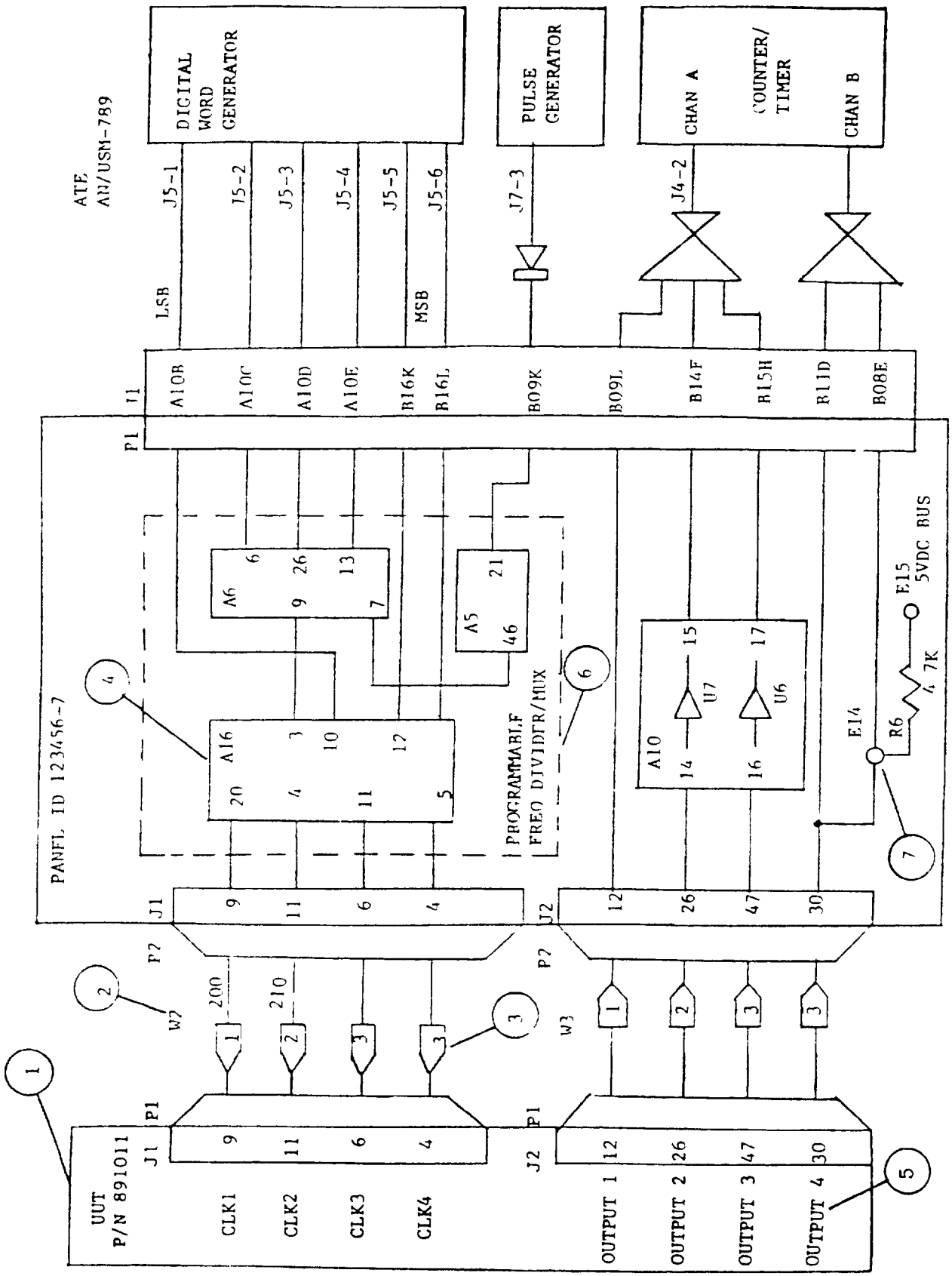


FIGURE 1 EXAMPLE OF TEST DIAGRAM

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